

SORBONNE UNIVERSITÉ

DEVELOPMENT AND VALIDATION OF COMPUTATIONAL TOOLS FOR  
AERODYNAMIC AND AEROELASTIC APPLICATIONS

VALENTIN COSSIN  
Master 1 in Mechanics  
Program: Fluid Mechanics

**MASTER'S INTERNSHIP REPORT**

submitted in partial fulfillment of the requirements for validation of the Master 1 degree

Host institution: Polytechnique Montréal  
Joseph-Armand Bombardier Institute  
Supervisor: Éric Laurendeau

August 2026

## ACKNOWLEDGMENTS

I would first like to express my sincere thanks to my supervisor, *Éric Laurendeau*, for giving me the opportunity to join his laboratory. I am also grateful for his financial support, which made my stay abroad possible, and for his attentive supervision and teaching throughout this work. His constant optimism about the project was a valuable source of confidence and motivation.

I also thank *Rawi* for his help at key stages of the Euler solver development. His code, used as a reference, saved considerable time, particularly during the initial implementation of normal, surface, and volume calculations and, at the end of the project, during the implementation of the multigrid method.

I am grateful to *Samuel Ayala* for providing the existing TinyVLM framework, including the low-fidelity aerodynamic methods and full-span aeroelastic implementation that provided an important starting point for the present development. I also thank *Samuel F.* and *Baptiste* for the discussions that contributed to the development of my CFD culture. The lunch breaks with the laboratory members will probably be among the things I miss most.

Finally, I thank the teaching and administrative staff of Sorbonne Université and Polytechnique Montréal for their support and guidance.

## RÉSUMÉ

Ce mémoire présente le développement et la vérification de deux outils numériques pour des applications aérodynamiques et aéroélastiques. Le premier est un solveur bidimensionnel des équations d'Euler compressibles, fondé sur une discrétisation volumes finis centrée, une dissipation artificielle de Swanson–Tukel/JST et une intégration en pseudo-temps par Runge–Kutta. Des techniques d'accélération par pas de temps local, lissage résiduel implicite et multigrille FAS/FMG sont également étudiées. La stratégie de vérification combine des tests élémentaires, des critères de résidu et des campagnes de raffinement de maillage pour des écoulements subsoniques et transsoniques, portants ou non portants. Les résultats distinguent explicitement convergence du résidu, stabilité des coefficients, convergence en maillage et régime asymptotique. Les quatre campagnes atteignent désormais  $2048^2$ . Les coefficients des cas subsoniques fins sont stabilisés malgré des résidus fins et oscillants, tandis que les cas transsoniques à  $2048^2$  sont pleinement convergés en résidu ; le triplet de maillages fins du cas transsonique portant ne relève toutefois pas d'un régime asymptotique à un seul terme.

Le second outil modélise une configuration aéroélastique à trois degrés de liberté associée à un volet muni d'un absorbeur d'énergie non linéaire. Le modèle couple une méthode des réseaux tourbillonnaires instationnaires à une intégration structurelle implicite. Les benchmarks statiques et instationnaires fondés sur la théorie des profils minces et sur la référence de Theodorsen permettent de vérifier la circulation, la réponse d'un volet articulé et les réponses harmoniques et biharmoniques. La comparaison avec les mesures rapportées par García Pérez et al. porte ensuite sur les fréquences structurelles, les retours libres et les réponses de grande amplitude. Les calculs UVLM convergés restent amortis et ne reproduisent pas de branche stable d'oscillation en cycle limite. Cette limitation est cohérente avec un modèle potentiel, inviscide et incompressible qui ne contient ni écoulement décollé ni modèle de décrochage dynamique ; elle ne suffit toutefois pas à expliquer à elle seule l'absence du seuil de flutter de faible amplitude.

## ABSTRACT

This thesis presents the development and verification of two computational tools for aerodynamic and aeroelastic applications. The first is a two-dimensional compressible Euler solver based on a centred finite-volume discretization, Swanson–Tukel/JST artificial dissipation, and pseudo-time Runge–Kutta integration. Local time stepping, implicit residual smoothing, and FAS/FMG multigrid acceleration are also considered. The verification strategy combines component tests, residual criteria, and grid-refinement campaigns for subsonic and transonic lifting and non-lifting flows. The results distinguish residual convergence, coefficient stability, grid convergence, and asymptotic behaviour. All four campaigns now extend to  $2048^2$ . The finest subsonic coefficients are stable despite finite oscillatory residual levels, whereas the transonic  $2048^2$  runs are fully residual-converged; the finest transonic lifting triplet is not a single-term asymptotic sequence.

The second tool models a three-degree-of-freedom aeroelastic configuration with a flap equipped with a nonlinear energy sink. The model couples an unsteady vortex-lattice method to an implicit structural integrator. Static and unsteady benchmarks based on thin-airfoil theory and the Theodorsen reference assess circulation, articulated-flap response, and harmonic and biharmonic motion. Comparisons with the measurements reported by García Pérez et al. then address structural frequencies, free-return responses, and finite-amplitude motion. The converged UVLM calculations remain damped and do not produce a stable limit-cycle branch. This limitation is consistent with a potential, inviscid, incompressible model without separated-flow or dynamic-stall modelling; it does not, however, by itself explain the missing low-amplitude flutter onset.

## TABLE OF CONTENTS

|                                                                    |      |
|--------------------------------------------------------------------|------|
| ACKNOWLEDGMENTS . . . . .                                          | ii   |
| RÉSUMÉ . . . . .                                                   | iii  |
| ABSTRACT . . . . .                                                 | iv   |
| TABLE OF CONTENTS . . . . .                                        | v    |
| LIST OF TABLES . . . . .                                           | viii |
| LIST OF FIGURES . . . . .                                          | ix   |
| LIST OF ACRONYMS AND ABBREVIATIONS . . . . .                       | x    |
| NOMENCLATURE . . . . .                                             | xi   |
| LIST OF APPENDICES . . . . .                                       | xii  |
| CHAPTER 1 INTRODUCTION . . . . .                                   | 1    |
| 1.1 Basic Concepts . . . . .                                       | 1    |
| 1.2 Elements of the Problematics . . . . .                         | 1    |
| 1.3 Objectives . . . . .                                           | 1    |
| 1.4 Plan of the Report . . . . .                                   | 1    |
| CHAPTER 2 TWO-DIMENSIONAL COMPRESSIBLE EULER SOLVER . . . . .      | 2    |
| 2.1 Governing Equations . . . . .                                  | 2    |
| 2.2 Finite-Volume Spatial Discretization . . . . .                 | 3    |
| 2.3 Central Scheme and JST Artificial Dissipation . . . . .        | 3    |
| 2.4 Boundary Conditions . . . . .                                  | 5    |
| 2.5 Far-Field State for Lifting Calculations . . . . .             | 5    |
| 2.6 Pseudo-Time Integration and Convergence Acceleration . . . . . | 6    |
| 2.6.1 RK5 and Local Time Stepping . . . . .                        | 6    |
| 2.6.2 Implicit Residual Smoothing . . . . .                        | 6    |
| 2.6.3 FAS Multigrid and FMG Initialization . . . . .               | 6    |
| 2.7 Aerodynamic Coefficients . . . . .                             | 7    |
| 2.8 Verification Strategy . . . . .                                | 8    |

|           |                                                                            |    |
|-----------|----------------------------------------------------------------------------|----|
| CHAPTER 3 | COMPRESSIBLE EULER SOLVER: NUMERICAL RESULTS . . .                         | 10 |
| 3.1       | Numerical Results and Grid Convergence . . . . .                           | 10 |
| 3.1.1     | Verification and Grid-Convergence Setup . . . . .                          | 10 |
| 3.1.2     | Solver Performance . . . . .                                               | 10 |
| 3.1.3     | Subsonic Results . . . . .                                                 | 11 |
| 3.1.4     | Transonic Results . . . . .                                                | 12 |
| 3.1.5     | Finest-grid summary and Vassberg–Jameson comparison . . . . .              | 13 |
| 3.2       | Limitations and Numerical Interpretation . . . . .                         | 15 |
| CHAPTER 4 | AEROELASTIC MODELLING AND SOLVER METHODOLOGY .                             | 17 |
| 4.1       | Aeroelastic Problem and Modelling Strategy . . . . .                       | 17 |
| 4.1.1     | Selection of the aerodynamic descriptions . . . . .                        | 17 |
| 4.2       | Michon Experimental Configuration . . . . .                                | 18 |
| 4.2.1     | Parameter provenance . . . . .                                             | 19 |
| 4.3       | Three-Degree-of-Freedom Structural Model . . . . .                         | 20 |
| 4.3.1     | Dimensional equations and generalized coordinates . . . . .                | 20 |
| 4.3.2     | Nondimensional form . . . . .                                              | 21 |
| 4.4       | Unsteady Aerodynamic Models . . . . .                                      | 21 |
| 4.4.1     | Jones realization of Theodorsen aerodynamics . . . . .                     | 21 |
| 4.4.2     | Multi-surface UVLM formulation . . . . .                                   | 22 |
| 4.4.3     | Michon multi-surface kinematics . . . . .                                  | 23 |
| 4.5       | Aerodynamic Forces and Generalized Loads . . . . .                         | 24 |
| 4.6       | Time Integration and Implicit Fluid–Structure Coupling . . . . .           | 25 |
| 4.6.1     | Newmark structural update . . . . .                                        | 25 |
| 4.6.2     | Partitioned implicit algorithm . . . . .                                   | 26 |
| CHAPTER 5 | AEROELASTIC NUMERICAL RESULTS . . . . .                                    | 28 |
| 5.1       | Aerodynamic Verification and Reference-Model Validation . . . . .          | 28 |
| 5.1.1     | Quasi-two-dimensional steady benchmarks . . . . .                          | 28 |
| 5.1.2     | Harmonic unsteady response . . . . .                                       | 30 |
| 5.1.3     | Biharmonic unsteady response . . . . .                                     | 32 |
| 5.2       | Application-Level Comparison with García Pérez et al.’s Measurements . . . | 33 |
| 5.2.1     | Zero-flow structural response . . . . .                                    | 34 |
| 5.2.2     | Linear aeroelastic response: Fig. 7 . . . . .                              | 34 |
| 5.2.3     | Nonlinear response: Fig. 8 . . . . .                                       | 36 |
| 5.2.4     | Scope and limitations . . . . .                                            | 37 |

|                                                                   |    |
|-------------------------------------------------------------------|----|
| CHAPTER 6 CONCLUSION . . . . .                                    | 38 |
| 6.1 Synthesis of Work . . . . .                                   | 38 |
| 6.2 Limitations of the Proposed Solution . . . . .                | 38 |
| 6.3 Future Work . . . . .                                         | 38 |
| REFERENCES . . . . .                                              | 39 |
| APPENDICES . . . . .                                              | 40 |
| A.1 Finite-Volume Geometry and Boundary Details . . . . .         | 40 |
| A.2 JST / Swanson–Tukel Boundary and Stencil Details . . . . .    | 40 |
| A.3 FAS / FMG Transfer Operators and Cycle Details . . . . .      | 42 |
| A.4 Component Verification Tests . . . . .                        | 43 |
| A.5 Complete Grid-Convergence Data . . . . .                      | 44 |
| A.6 Iterative Convergence and Solver Performance . . . . .        | 46 |
| B.1 Structural scaling and parameter provenance . . . . .         | 47 |
| B.2 Jones–Theodorsen matrix realization . . . . .                 | 49 |
| B.3 UVLM discrete assembly and wake bookkeeping . . . . .         | 50 |
| B.3.1 Hierarchical rigid-body kinematics . . . . .                | 51 |
| B.4 Newmark- $\beta$ integration . . . . .                        | 53 |
| B.4.1 Kinematic assumptions . . . . .                             | 53 |
| B.4.2 Effective structural system . . . . .                       | 53 |
| B.4.3 Incremental form used by the solver . . . . .               | 54 |
| B.5 Inferred-balanced structural reconstruction . . . . .         | 55 |
| B.6 Linear response and frequency extraction . . . . .            | 55 |
| B.7 Nonlinear response protocol . . . . .                         | 56 |
| B.8 Calibration and independence of the validation data . . . . . | 56 |
| B.9 Experimental-data and model limitations . . . . .             | 57 |

## LIST OF TABLES

|     |                                                                                                                                                                                                                             |    |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 3.1 | Finest-grid orders and reference comparisons. A dash indicates that the corresponding sequence or reference limit is not applicable. . . . .                                                                                | 13 |
| 4.1 | Geometric quantities used for the Michon configuration. Values in the first group are reported by García Pérez et al.; the last two quantities are direct geometric conversions used by the present mesh generator. . . . . | 18 |
| 5.1 | Harmonic prescribed-pitch cases and relative RMS errors. . . . .                                                                                                                                                            | 31 |
| 5.2 | Biharmonic prescribed-pitch cases and component-wise errors. . . . .                                                                                                                                                        | 32 |
| 5.3 | Zero-flow frequencies of the inferred-balanced structural reconstruction compared with approximate readings of Michon Fig. 7. . . . .                                                                                       | 34 |
| 5.4 | UVLM free-return frequencies for the Fig. 7 comparison. . . . .                                                                                                                                                             | 35 |
| A.1 | Subsonic nonlifting grid coefficients . . . . .                                                                                                                                                                             | 44 |
| A.2 | Subsonic lifting grid coefficients . . . . .                                                                                                                                                                                | 45 |
| A.3 | Transonic nonlifting grid coefficients . . . . .                                                                                                                                                                            | 45 |
| A.4 | Transonic lifting grid coefficients . . . . .                                                                                                                                                                               | 45 |
| B.1 | Provenance of structural quantities used by the Michon path. . . . .                                                                                                                                                        | 48 |

## LIST OF FIGURES

|     |                                                                                                                                                                                                                          |    |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 2.1 | Structured O-grid around the NACA0012 profile. . . . .                                                                                                                                                                   | 3  |
| 2.2 | Pressure sensor and JST dissipation stencil. . . . .                                                                                                                                                                     | 5  |
| 2.3 | Schematic of the FAS multigrid hierarchy and W-cycle. High-frequency error is efficiently damped on each level, while low-frequency fine-grid error becomes easier to reduce after restriction to coarser grids. . . . . | 7  |
| 3.1 | Residual convergence for the $32 \times 32$ benchmark using single-grid RK5 and FAS multigrid. . . . .                                                                                                                   | 11 |
| 3.2 | Pressure field (left) and Mach-number field (right) for the $2048 \times 2048$ transonic nonlifting case, $M_\infty = 0.80$ , $\alpha = 0^\circ$ . . . . .                                                               | 12 |
| 3.3 | Grid-convergence errors $ C_i - C_i^* $ versus $h = 1/N$ for the four Euler benchmark cases. Filled symbols indicate the grid sequence used for the reported positive-order estimates. . . . .                           | 14 |
| 3.4 | Subsonic lifting surface-pressure comparison . . . . .                                                                                                                                                                   | 15 |
| 4.1 | Geometry and generalized coordinates of the Michon flap–NES configuration. . . . .                                                                                                                                       | 19 |
| 4.2 | Michon UVLM multi-surface discretization in the undeformed configuration. The bound lattice comprises the full-span forward surface, two aft outboard surfaces, and the central flap surface. . . . .                    | 23 |
| 5.1 | Static $A_0$ – $A_1$ reconstruction benchmark. . . . .                                                                                                                                                                   | 29 |
| 5.2 | Static flap-effect benchmark: sectional and global UVLM lift responses compared with the thin-airfoil flap reference. . . . .                                                                                            | 30 |
| 5.3 | Harmonic prescribed-pitch benchmark. . . . .                                                                                                                                                                             | 31 |
| 5.4 | Biharmonic prescribed-pitch benchmark. . . . .                                                                                                                                                                           | 33 |
| 5.5 | Sparse UVLM frequency comparison for Fig. 7 of García Pérez et al. . . . .                                                                                                                                               | 35 |
| 5.6 | UVLM comparison for Fig. 8 of García Pérez et al. . . . .                                                                                                                                                                | 36 |
| A.1 | Single-thread and four-thread timing comparison for the $512^2$ case. . . . .                                                                                                                                            | 46 |
| B.1 | Calibrated response based on García Pérez et al. Fig. 7. . . . .                                                                                                                                                         | 57 |

## LIST OF ACRONYMS AND ABBREVIATIONS

|      |                                               |
|------|-----------------------------------------------|
| CFD  | Computational Fluid Dynamics                  |
| CFL  | Courant-Friedrichs-Lewy number                |
| FAS  | Full Approximation Scheme                     |
| FMG  | Full Multigrid                                |
| FSI  | Fluid–Structure Interaction                   |
| IRS  | Implicit Residual Smoothing                   |
| JST  | Jameson–Schmidt–Turbel artificial dissipation |
| NES  | Nonlinear Energy Sink                         |
| RK   | Runge–Kutta                                   |
| UVLM | Unsteady Vortex Lattice Method                |
| VLM  | Vortex Lattice Method                         |

## NOMENCLATURE

| Symbol                                     | Meaning                                                                       |
|--------------------------------------------|-------------------------------------------------------------------------------|
| $h$                                        | Plunge displacement.                                                          |
| $\alpha$                                   | Main-body pitch angle about the elastic axis.                                 |
| $\beta$                                    | Flap rotation relative to the main body.                                      |
| $\mathbf{q}$                               | Dimensional generalized-coordinate vector.                                    |
| $\mathbf{x}$                               | Dimensionless generalized-coordinate vector.                                  |
| $\mathbf{M}_s, \mathbf{C}_s, \mathbf{K}_s$ | Structural mass, damping, and stiffness matrices in the normalized equations. |
| $\mathbf{Q}_a$                             | Generalized aerodynamic-load vector.                                          |
| $\rho$                                     | Fluid density.                                                                |
| $U$                                        | Freestream velocity.                                                          |
| $b$                                        | Airfoil semi-chord.                                                           |
| $k$                                        | Reduced frequency.                                                            |
| $\mu$                                      | Mass ratio used by the reduced aeroelastic model.                             |
| $\mathbf{\Gamma}_b$                        | Bound circulation vector.                                                     |
| $\mathbf{\Gamma}_w$                        | Wake circulation vector.                                                      |
| $\mathbf{X}_w$                             | Wake-vertex coordinates.                                                      |
| $\mathbf{n}$                               | Panel normal vector.                                                          |
| $\mathbf{A}$                               | UVLM influence matrix.                                                        |
| $\Delta\tau$                               | Dimensionless time step.                                                      |
| $\beta_N, \gamma_N$                        | Newmark parameters.                                                           |
| $\boldsymbol{\lambda}$                     | Jones–Theodorsen aerodynamic lag-state vector.                                |
| $\mathbf{T}$                               | Homogeneous kinematic transformation matrix.                                  |
| $\mathbf{R}$                               | Rotational block of a homogeneous transformation.                             |
| $\boldsymbol{\tau}$                        | Translation block of a homogeneous transformation.                            |

**LIST OF APPENDICES**

|            |                                                    |    |
|------------|----------------------------------------------------|----|
| Appendix A | EULER SOLVER TECHNICAL AND NUMERICAL DETAILS . . . | 40 |
| Appendix B | AEROELASTIC TECHNICAL DETAILS . . . . .            | 47 |

## CHAPTER 1 INTRODUCTION

Computational aerodynamic models span a hierarchy of physical fidelity and cost. This report develops and verifies tools across that hierarchy, with the principal scientific focus on low-fidelity aeroelastic modelling.

### 1.1 Basic Concepts

Lower-fidelity models enable repeated preliminary-design and parametric studies. Aeroelasticity couples structural motion and aerodynamic loads; verification concerns the numerical model, whereas validation concerns physical measurements. Flutter and LCOs are considered in this sense.

### 1.2 Elements of the Problematics

High-fidelity CFD resolves detailed physics but can be too costly for repeated early-design and aeroelastic load calculations. The industrial load-process perspective described by Garrigues motivates efficient aerodynamic and structural reductions for broad configuration studies, while higher-fidelity calculations or experiments remain reference cases (Garrigues, 2018).

The scientific project models a flexible wing with an articulated flap as a possible nonlinear energy sink (NES), using the partial-span configuration studied experimentally by García Pérez et al. The Euler-solver work preceded it as a pedagogical numerical-development exercise.

### 1.3 Objectives

The internship had two objectives: develop and verify a two-dimensional compressible Euler solver as an autonomous CFD exercise; then develop and critically assess a low-fidelity aeroelastic model for the partial-span flap–NES configuration against analytical/reference results and the experiments reported by García Pérez et al.

### 1.4 Plan of the Report

Chapters 2–3: Euler; Chapters 4–5: aeroelasticity; Chapter 6: conclusions; appendices: technical details.

## CHAPTER 2 TWO-DIMENSIONAL COMPRESSIBLE EULER SOLVER

This chapter describes the two-dimensional compressible Euler solver. It is a cell-centred finite-volume method on a structured O-mesh. The spatial discretization combines a centred flux with Swanson–Tukel/JST artificial dissipation. Steady solutions are obtained with a five-stage Runge–Kutta scheme, local time stepping, optional implicit residual smoothing (IRS), and optional FAS multigrid with FMG initialization.

### 2.1 Governing Equations

The flow is modelled as inviscid and compressible. In Cartesian coordinates, the conservative Euler equations are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} = 0, \quad (2.1)$$

with

$$\mathbf{U} = [\rho \quad \rho u \quad \rho v \quad \rho E]^T, \quad \mathbf{W} = [\rho \quad u \quad v \quad p]^T. \quad (2.2)$$

Here,  $\rho$  is density,  $(u, v)$  is velocity,  $p$  is pressure, and  $E$  is specific total energy. The Cartesian fluxes are

$$\mathbf{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (\rho E + p)u \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ (\rho E + p)v \end{bmatrix}. \quad (2.3)$$

The perfect-gas closure is

$$E = \frac{p}{(\gamma - 1)\rho} + \frac{u^2 + v^2}{2}, \quad a = \sqrt{\frac{\gamma p}{\rho}}, \quad \gamma = 1.4, \quad (2.4)$$

where  $a$  is the speed of sound. States with non-positive density or pressure are rejected. For a face with outward unit normal  $\mathbf{n} = (n_x, n_y)$ ,

$$u_n = un_x + vn_y, \quad \mathbf{F}_n = n_x \mathbf{F} + n_y \mathbf{G} = \begin{bmatrix} \rho u_n \\ \rho u u_n + p n_x \\ \rho v u_n + p n_y \\ (\rho E + p)u_n \end{bmatrix}. \quad (2.5)$$

The pseudo-time residual is driven to the selected convergence criterion using local stepping and, when enabled, multigrid acceleration (Blazek, 2005; Swanson and Turkel, 1997).

## 2.2 Finite-Volume Spatial Discretization

The structured quadrilateral O-mesh stores the conservative state at cell centres. Figure 2.1 shows its topology: the circumferential direction is periodic, while the radial direction runs from the airfoil to the far-field boundary. Cell areas, centroids, outward face vectors, and the bounded index conventions are defined in Appendix A.1.

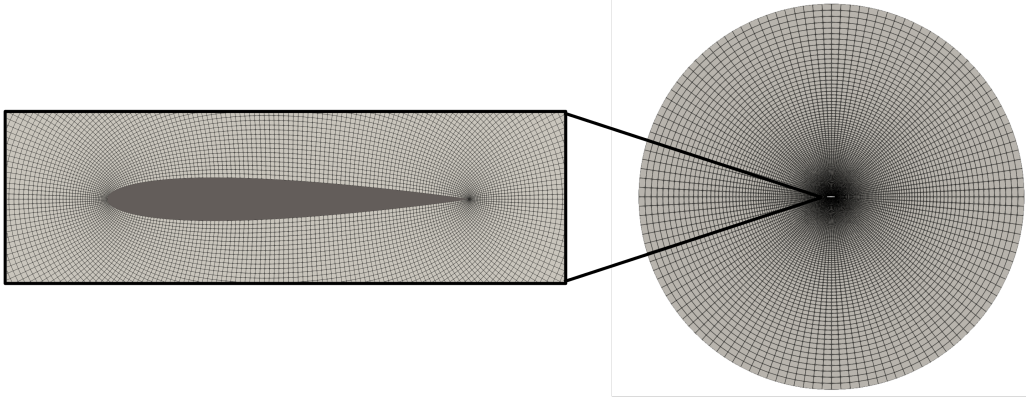


Figure 2.1 Structured O-grid around the NACA0012 profile.

The semi-discrete finite-volume residual is

$$\mathbf{R}_{ij}(\mathbf{U}) = \frac{1}{\Omega_{ij}} \sum_{f \in \partial C_{ij}} \sigma_{ij,f} \Phi_f, \quad \frac{d\mathbf{U}_{ij}}{dt} + \mathbf{R}_{ij} = 0, \quad (2.6)$$

where  $\sigma_{L,f} = +1$  and  $\sigma_{R,f} = -1$  for an interior face oriented from left to right. The circumferential index is periodic and the radial index is bounded.

## 2.3 Central Scheme and JST Artificial Dissipation

The centred convective flux is

$$\Phi_f^c = \frac{\ell_f}{2} [\mathbf{F}_n(\mathbf{U}_L) + \mathbf{F}_n(\mathbf{U}_R)]. \quad (2.7)$$

The Swanson–Turkel/JST dissipation is added to this centred flux without reconstruction or a slope limiter. It combines a pressure sensor, second-difference dissipation near sharp pressure changes, and a fourth- difference term in smooth regions. The pressure sensor and

face maximum are

$$\nu_r = \frac{|p_{r-1} - 2p_r + p_{r+1}|}{|p_{r-1}| + 2|p_r| + |p_{r+1}|}, \quad m_{r+1/2} = \max_{q \in \{r-1, r, r+1, r+2\}} \nu_q. \quad (2.8)$$

The coefficients are

$$\varepsilon_{r+1/2}^{(2)} = k_2 m_{r+1/2}, \quad \varepsilon_{r+1/2}^{(4)} = \max\left(0, k_4 - \varepsilon_{r+1/2}^{(2)}\right), \quad k_2 = \frac{1}{4}, \quad k_4 = \frac{1}{32}. \quad (2.9)$$

With

$$\Delta \mathbf{U}_{r+1/2} = \mathbf{U}_{r+1} - \mathbf{U}_r, \quad \Delta^3 \mathbf{U}_{r+1/2} = \mathbf{U}_{r+2} - 3\mathbf{U}_{r+1} + 3\mathbf{U}_r - \mathbf{U}_{r-1}, \quad (2.10)$$

the integrated dissipative flux is

$$\mathbf{Q}_{r+1/2} = \Lambda_{r+1/2,d} \left[ \varepsilon_{r+1/2}^{(2)} \Delta \mathbf{U}_{r+1/2} - \varepsilon_{r+1/2}^{(4)} \Delta^3 \mathbf{U}_{r+1/2} \right]. \quad (2.11)$$

For direction  $d \in \{i, j\}$ , with  $d^\perp$  denoting the transverse direction, the present solver uses the scalar, transverse-coupled scaling

$$\lambda_{C,d} = (|\mathbf{u}_C \cdot \mathbf{n}_{C,d}| + a_C) \ell_{C,d}, \quad \Lambda_{C,d} = \lambda_{C,d} + \sqrt{\lambda_{C,d} \lambda_{C,d^\perp}}, \quad \Lambda_{r+1/2,d} = \frac{1}{2} (\Lambda_{r,d} + \Lambda_{r+1,d}). \quad (2.12)$$

Here  $|\mathbf{u}_C \cdot \mathbf{n}_{C,d}| + a_C$  is the local scalar spectral-radius estimate in direction  $d$ , and  $\ell_{C,d}$  supplies the face measure. The transverse contribution retains sensitivity to the second grid direction, which is useful for dissipation scaling on curvilinear or anisotropic meshes. This is a scalar multiplier on the state differences, not a characteristic matrix dissipation; the detailed stencil is given in Appendix A.2 and follows the Swanson–Tukel motivation for direction-aware scaling (Swanson and Turkel, 1997).

The total face flux is

$$\Phi_f = \Phi_f^c - \mathbf{Q}_f. \quad (2.13)$$

On multigrid coarse levels, a first-order dissipation branch is selected for robustness. The boundary closure and coarse-level stencil are given in Appendix A.2.

Figure 2.2 illustrates the pressure-sensor stencil and the associated difference operators. The second-difference contribution is activated near strong pressure variations, whereas the fourth-difference term provides the dominant background dissipation in smooth regions.

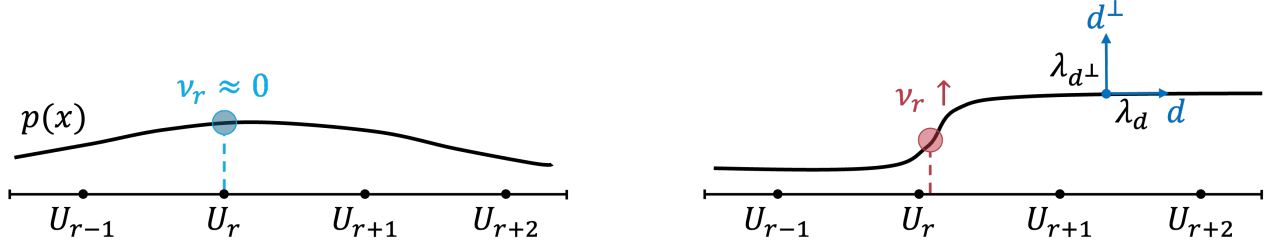


Figure 2.2 Pressure sensor and JST dissipation stencil.

## 2.4 Boundary Conditions

Two ghost layers are used. The update order is wall, far field, then periodic copying; corner values outside the physical index ranges are neither written nor read. The wall and far-field states are expressed in primitive variables.

Periodic ghost layers copy the corresponding physical states across the circumferential seam. At an inviscid wall, the normal velocity is reflected while density and pressure are preserved:

$$\rho_g = \rho_i, \quad p_g = p_i, \quad \mathbf{u}_g = \mathbf{u}_i - 2u_n \mathbf{n}_w. \quad (2.14)$$

The characteristic far field uses the local normal Mach regime: incoming characteristics are prescribed from the corrected free stream, while outgoing information is taken from the interior. The complete subsonic states, supersonic branches, wall pressure, and ghost-layer fallback are given in Appendix A.2.

## 2.5 Far-Field State for Lifting Calculations

The uncorrected free stream is defined from  $(M_\infty, \alpha)$  with  $\rho_\infty = 1$  and  $p_\infty = 1/\gamma$ :

$$u_\infty = M_\infty \sqrt{\gamma} \cos \alpha, \quad v_\infty = M_\infty \sqrt{\gamma} \sin \alpha. \quad (2.15)$$

When the optional point-vortex correction is enabled, with  $\Delta x = x - x_v$ ,  $\Delta y = y - y_v$ , and  $r^2 = \Delta x^2 + \Delta y^2$ ,

$$u_{\infty, \Gamma} = u_\infty - \frac{\Gamma \Delta y}{2\pi r^2}, \quad v_{\infty, \Gamma} = v_\infty + \frac{\Gamma \Delta x}{2\pi r^2}, \quad (2.16)$$

and

$$\begin{aligned} a_\Gamma^2 &= a_\infty^2 + \frac{\gamma-1}{2} (V_\infty^2 - V_{\infty, \Gamma}^2), & \theta_\Gamma &= \frac{a_\Gamma^2}{a_\infty^2}, \\ \rho_{\infty, \Gamma} &= \rho_\infty \theta_\Gamma^{1/(\gamma-1)}, & p_{\infty, \Gamma} &= p_\infty \theta_\Gamma^{\gamma/(\gamma-1)}, & \frac{p_{\infty, \Gamma}}{\rho_{\infty, \Gamma}^\gamma} &= \frac{p_\infty}{\rho_\infty^\gamma}. \end{aligned} \quad (2.17)$$

This preserves the total enthalpy while introducing the circulation field. The circulation target and relaxation are

$$\Gamma_n^* = -\frac{L_n}{\rho_\infty V_\infty}, \quad \Gamma_{n+1} = \Gamma_n + \omega_v(\Gamma_n^* - \Gamma_n), \quad \omega_v = 0.05, \quad (2.18)$$

with default vortex centre  $(x_v, y_v) = (0.25, 0)$ . A prescribed circulation disables relaxation. The correction acts on the far-field state, not by a direct relaxation of the complete flow state. The update timing and multigrid synchronization are specified in Appendix A.2.

## 2.6 Pseudo-Time Integration and Convergence Acceleration

The local pseudo-time step is

$$\Delta t_{ij} = \text{CFL}_n \frac{\Omega_{ij}}{\sum_{f \in \partial C_{ij}} (|\mathbf{u}_{ij} \cdot \mathbf{n}_f| + a_{ij}) \ell_f}. \quad (2.19)$$

The CFL number may be fixed or ramped during the pseudo-time integration.

### 2.6.1 RK5 and Local Time Stepping

Let  $\mathbf{U}^{(0)} = \mathbf{U}^n$ . The five stages use

$$(\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5) = \left(\frac{1}{4}, \frac{1}{6}, \frac{3}{8}, \frac{1}{2}, 1\right), \quad \mathbf{U}^{(s)} = \mathbf{U}^{(0)} - \alpha_s \Delta t \widetilde{\mathbf{R}}^{(s)}. \quad (2.20)$$

At each stage, the ghost states and residual are recomputed, physicality is checked, and a forced FAS residual is smoothed when applicable. The detailed CFL schedule and stage ordering are given in Appendix A.3.

### 2.6.2 Implicit Residual Smoothing

IRS is an optional directional implicit filter applied to the pseudo-time residual before the Runge–Kutta update. It uses cyclic and bounded tridiagonal solves; its complete stencil is given in Appendix A.3.

### 2.6.3 FAS Multigrid and FMG Initialization

The hierarchy halves both mesh dimensions. FAS transfers the nonlinear state and residual between levels rather than solving a separately linearized error equation. With  $h$  and  $H$

denoting fine and coarse levels, respectively,

$$\mathbf{U}^H = R_A \mathbf{U}^h, \quad \mathbf{r}^h = L^h(\mathbf{U}^h) - \mathbf{f}^h, \quad \mathbf{f}^H = L^H(R_A \mathbf{U}^h) - R_A \mathbf{r}^h, \quad L^H(\mathbf{U}^H) = \mathbf{f}^H. \quad (2.21)$$

After the forced coarse problem has been smoothed, its correction is prolonged to the fine level as

$$\mathbf{U}^h \leftarrow \mathbf{U}^h + \omega_\ell P \left( \mathbf{U}^H - \mathbf{U}_{\text{pre}}^H \right). \quad (2.22)$$

Runge–Kutta smoothing efficiently damps high-frequency error on a given grid, whereas low-frequency error is difficult to remove on the fine grid. After restriction, the same error is represented at a higher relative frequency on the coarse grid and is more effectively smoothed; prolongation then returns the correction to the fine level. A W-cycle revisits coarse levels more often than a V-cycle, while FMG constructs a coarse-to-fine initial approximation before the final fine-grid FAS iterations. Figure 2.3 illustrates this hierarchy and W-cycle mechanism. Exact transfer weights, index handling, dissipation choices, and cycle bookkeeping are given in Appendix A.3. These multistage, residual-smoothing, and multigrid strategies are described by Swanson and Turkel (Swanson and Turkel, 1997).

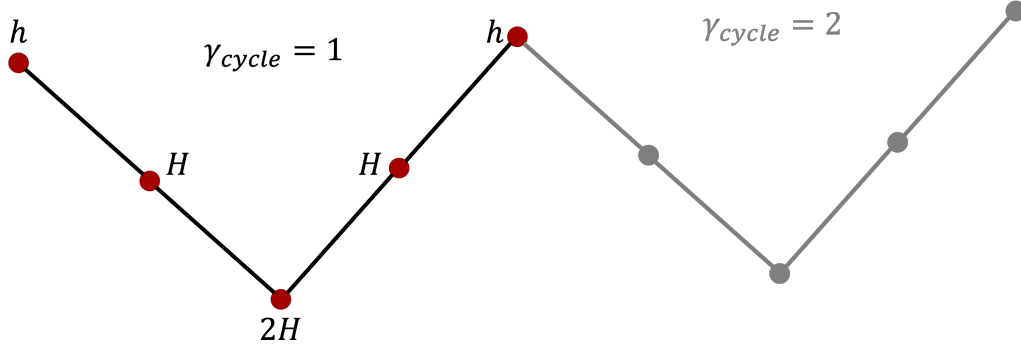


Figure 2.3 Schematic of the FAS multigrid hierarchy and W-cycle. High-frequency error is efficiently damped on each level, while low-frequency fine-grid error becomes easier to reduce after restriction to coarser grids.

## 2.7 Aerodynamic Coefficients

The wall pressure is extrapolated from the first two cell-centre states using the wall-normal distances defined in Section 2.4. The pressure force per unit span is integrated with the outward normal of the fluid cell,

$$\mathbf{F}_p = \sum_{f \in \partial\Omega_w} p_{w,f} \mathbf{n}_f \ell_f, \quad F_x = \sum_f p_{w,f} n_{x,f} \ell_f, \quad F_y = \sum_f p_{w,f} n_{y,f} \ell_f. \quad (2.23)$$

The free-stream pressure is not subtracted in this force integral. With  $D = F_x \cos \alpha + F_y \sin \alpha$  and  $L = -F_x \sin \alpha + F_y \cos \alpha$ ,

$$q_\infty = \frac{1}{2} \rho_\infty V_\infty^2, \quad L_{ref} = 1, \quad C_D = \frac{D}{q_\infty L_{ref}}, \quad C_L = \frac{L}{q_\infty L_{ref}}. \quad (2.24)$$

The code moment about  $(x_{ref}, y_{ref}) = (0.25, 0)$  is

$$M_{code} = - \sum_f [(x_f - x_{ref}) dF_y - (y_f - y_{ref}) dF_x], \quad C_M = \frac{M_{code}}{q_\infty L_{ref}^2}, \quad (2.25)$$

where  $dF_x = p_{w,f} n_{x,f} \ell_f$  and  $dF_y = p_{w,f} n_{y,f} \ell_f$ . The surface pressure coefficient is

$$C_{p,f} = \frac{p_{w,f} - p_\infty}{q_\infty}. \quad (2.26)$$

## 2.8 Verification Strategy

Verification separates component checks, residual stopping, and grid convergence. The executed checks and the tests not claimed as completed are listed in Appendix A.4. The automated Euler test suite passes.

The residual norm is

$$\|\mathbf{R}\|_\infty = \max_{\substack{i,j \\ m \in \{\rho, \rho u, \rho v, \rho E\}}} |R_{ij,m}|. \quad (2.27)$$

For a mono-grid run, the stopping test is based on the logarithm of the residual reduction relative to its initial value. For multigrid, the criterion is  $\|\mathbf{R}^n\|_\infty \leq 10^{-6}$ ; no stopping criterion is based on changes in  $C_L$ ,  $C_D$ , or  $C_M$ . The implementation-level floor used by the mono-grid expression is given in Appendix A.4.

The reported grid family contains  $N \times N$  cells with  $N \in \{32, 64, 128, 256, 512, 1024, 2048\}$  for all four campaigns. For three consecutive grids, with  $N_m = 2N_c$  and  $N_f = 2N_m$ , the observed order is

$$p_{3g}(Q) = \log_2 \left| \frac{Q_{N_m} - Q_{N_c}}{Q_{N_f} - Q_{N_m}} \right|. \quad (2.28)$$

When a continuum value  $Q^*$  is available, the corresponding known-limit order is

$$p_{N \rightarrow 2N}^*(Q) = \log_2 \left| \frac{Q_N - Q^*}{Q_{2N} - Q^*} \right|. \quad (2.29)$$

For steady subcritical inviscid flow,  $C_D^* = 0$ , so  $p_{N \rightarrow 2N} = \log_2 [C_D(N)/C_D(2N)]$  when the ratio is positive. The grid-convergence interpretation follows Vassberg and Jameson (Vassberg and

Jameson, 2010); comparisons must match geometry, Mach number, angle of attack, far-field treatment, reference chord, force signs, and convergence criteria.

## CHAPTER 3 COMPRESSIBLE EULER SOLVER: NUMERICAL RESULTS

### 3.1 Numerical Results and Grid Convergence

The four campaigns combine subsonic or transonic flow with nonlifting or lifting conditions. The coefficients are computed from the extrapolated wall pressure described in Section 2.7; the complete coefficient sequences are reported in Appendix A.5. The comparison with Vassberg–Jameson is a CFD benchmark comparison, not an experimental validation (Vassberg and Jameson, 2010).

Four notions are distinguished. Iterative convergence concerns the reduction of the nonlinear or pseudo-time residual on one fixed grid. Coefficient stability concerns the change in  $C_L$ ,  $C_D$ , and  $C_M$  during the final iterations of that run. Grid convergence concerns the variation of a coefficient under mesh refinement. Asymptotic convergence is a stronger condition: the successive grid errors must be dominated by one leading power-law term before a Richardson continuum extrapolation is meaningful. Consequently, a stable coefficient on a fine grid is not automatically a residual-converged solution, and an observed three-grid order is not automatically an asymptotic order.

#### 3.1.1 Verification and Grid-Convergence Setup

The grid family contains  $N \times N$  finite-volume cells with  $N = 2^j$ ,  $j = 5, \dots, 11$  ( $32^2$  to  $2048^2$ ). The four campaigns therefore include the finest grid for both subsonic and transonic, lifting and nonlifting flow. The three-grid orders below use the consecutive-grid definition given in Section 2.8. When a published or exact continuum value is available, the known-limit order is reported separately.

#### 3.1.2 Solver Performance

Figure 3.1 compares the residual histories for the  $32 \times 32$  benchmark case using single-grid and FAS multigrid convergence. A substantial acceleration is already visible on this smallest grid of the refinement family. This is an iterative acceleration result and does not by itself establish grid or coefficient convergence. The separate one- versus four-thread timing comparison is retained in Appendix A.6 as an implementation-level scaling result, not as numerical verification evidence.

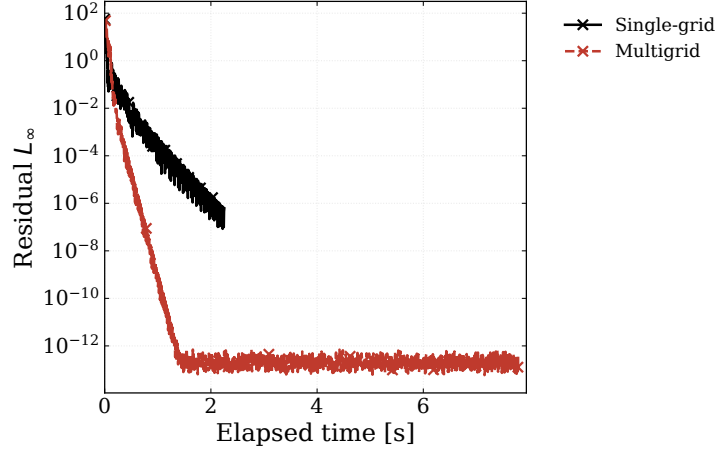


Figure 3.1 Residual convergence for the  $32 \times 32$  benchmark using single-grid RK5 and FAS multigrid.

### 3.1.3 Subsonic Results

For the subsonic nonlifting case ( $M_\infty = 0.50$ ,  $\alpha = 0^\circ$ ), the complete coefficient sequence is given in Appendix A.5.

$C_L$  and  $C_M$  remain at the numerical symmetry floor, while the drag approaches the exact inviscid subcritical limit  $C_D^* = 0$ . The  $2048^2$  coefficients are stable, but the residual remains finite and oscillatory; this point is therefore used for coefficient-based grid convergence and is not described as fully residual-converged.

For the finest triplet (512, 1024, 2048),  $p_{3g}(C_D) = 1.7236$ . The exact-limit order for the  $1024 \rightarrow 2048$  refinement is 1.8882. No order is assigned to  $C_L$  or  $C_M$  because they remain at the symmetry floor. These drag orders indicate an approach toward second-order behaviour over the finest grids, not a globally second-order sequence.

For the subsonic lifting case ( $M_\infty = 0.50$ ,  $\alpha = 1.25^\circ$ ), the complete coefficient sequence is given in Appendix A.5.

The published Vassberg–Jameson continuum limits are  $C_L^* = 0.180345850$ ,  $C_D^* = 0$ , and  $C_M^* = -0.002268812$  (Vassberg and Jameson, 2010). The corresponding three-grid and reference-limit orders are listed in Table 3.1. At  $2048^2$ , the coefficient discrepancies are 0.0461% for  $C_L$  and 0.605% for  $C_M$ ; the drag magnitude is  $|C_D| = 0.00982$  drag count. The coefficients are stable, but the residual remains finite and oscillatory. This point is therefore used for coefficient-based grid convergence, not as a fully residual-converged state.

### 3.1.4 Transonic Results

For transonic nonlifting flow ( $M_\infty = 0.80$ ,  $\alpha = 0^\circ$ ), the complete coefficient sequence is given in Appendix A.5.

The symmetry-floor behaviour of lift and moment persists through the finest grid. The  $2048^2$  run is fully residual-converged and coefficient-stable.

For the transonic nonlifting case, the two drag triplets give positive observed orders, as reported in Table 3.1; their Richardson estimates are  $C_D^{\text{Rich.}} = 0.0083443761$  and  $0.0083422664$ , respectively. No order is assigned to  $C_L$  or  $C_M$ , which remain at the symmetry floor. The  $2048^2$  result is used directly rather than inferred from the Richardson estimate.

Figure 3.2 shows the resulting approximately symmetric pressure and Mach fields. The symmetric compression pattern is consistent with the near-zero lift and pitching moment, while the transonic shocks still produce non-zero inviscid wave drag.

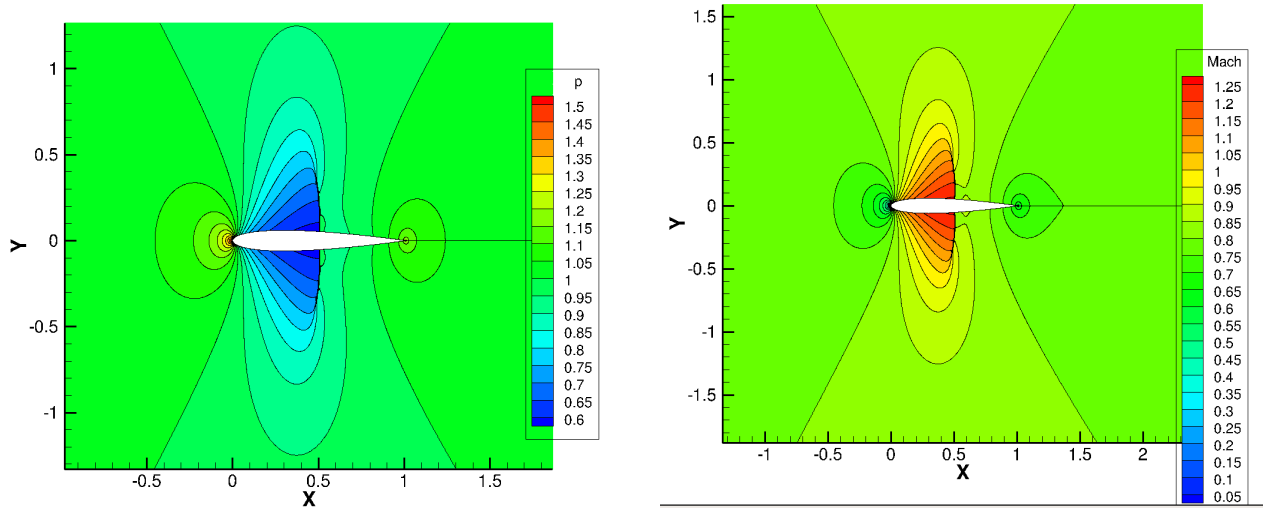


Figure 3.2 Pressure field (left) and Mach-number field (right) for the  $2048 \times 2048$  transonic nonlifting case,  $M_\infty = 0.80$ ,  $\alpha = 0^\circ$ .

For transonic lifting flow ( $M_\infty = 0.80$ ,  $\alpha = 1.25^\circ$ ), the complete coefficient sequence is given in Appendix A.5.

The coarser triplet is monotone and gives positive observed orders. The finest triplet gives negative orders for all three quantities (Table 3.1). These values describe finite-grid sequences only; they do not establish a globally second-order method. This triplet is not in a single-term asymptotic regime and must not be used for Richardson continuum extrapolation. Relative to the published continuum values, the discrepancies are 0.103%, 0.096%, and 0.400% for  $C_L$ ,  $C_D$ , and  $C_M$ , respectively. The run is fully residual-converged and coefficient-stable, but

the negative orders of the finest triplet preclude a single-term Richardson extrapolation.

### 3.1.5 Finest-grid summary and Vassberg–Jameson comparison

The published Vassberg–Jameson continuum values for the transonic lifting benchmark are (Vassberg and Jameson, 2010)

$$C_L^* = 0.356208937, \quad C_D^* = 0.022684938, \quad C_M^* = -0.038774022. \quad (3.1)$$

The subsonic lifting reference values used above are also published continuum values, namely  $C_L^* = 0.180345850$ ,  $C_D^* = 0$ , and  $C_M^* = -0.002268812$  (Vassberg and Jameson, 2010). The resulting finest-grid indicators are collected in Table 3.1. For the transonic nonlifting benchmark, the published continuum drag is  $C_D^* = 0.008342298$ ; the 2048<sup>2</sup> value is 0.0244% below this reference, with  $p_{1024 \rightarrow 2048}^* = 1.7374$ .

Table 3.1 Finest-grid orders and reference comparisons. A dash indicates that the corresponding sequence or reference limit is not applicable.

| Case                 | Quantity | $p_{3g}$     | $p_{3g}$      | $p_{1024 \rightarrow 2048}^*$ | finest error or Richardson value |
|----------------------|----------|--------------|---------------|-------------------------------|----------------------------------|
|                      |          | 256/512/1024 | 512/1024/2048 |                               |                                  |
| Subsonic nonlifting  | $C_D$    | –            | 1.7236        | 1.8882                        | –                                |
| Subsonic lifting     | $C_L$    | –            | 0.7443        | 0.7470                        | 0.0461%                          |
| Subsonic lifting     | $C_D$    | –            | 1.6796        | 2.3556                        | 0.00982 drag count               |
| Subsonic lifting     | $C_M$    | –            | 0.5771        | 0.9378                        | 0.605%                           |
| Transonic nonlifting | $C_D$    | 1.4890       | 1.7533        | 1.7374                        | 0.0244%                          |
| Transonic lifting    | $C_L$    | 1.4940       | −0.5877       | 2.5036                        | 0.103%                           |
| Transonic lifting    | $C_D$    | 1.4845       | −0.8372       | 2.3402                        | 0.096%                           |
| Transonic lifting    | $C_M$    | 1.4932       | −0.6061       | 2.0736                        | 0.400%                           |

For the transonic lifting case, the 2048<sup>2</sup> discrepancies and the reference-limit orders are those listed in Table 3.1. The run is fully residual-converged and coefficient-stable, but the negative orders of the finest triplet show that its grid sequence is not in a single-term asymptotic regime.

Open symbols denote retained grid data outside those sequences; in the transonic lifting panel, the 2048<sup>2</sup> point remains open because the finest triplet is non-asymptotic.

The integrated coefficient comparison is complemented by the local overlay in Fig. 3.4. The FLO82 curve is represented by digitized points from the published reference figure, whereas the red curve is computed by the present solver. The digitized FLO82 points correspond to the 4096 × 4096 solution shown in Fig. 14 of Vassberg and Jameson (Vassberg and Jameson, 2010); the comparison is therefore subject to finite digitization uncertainty.

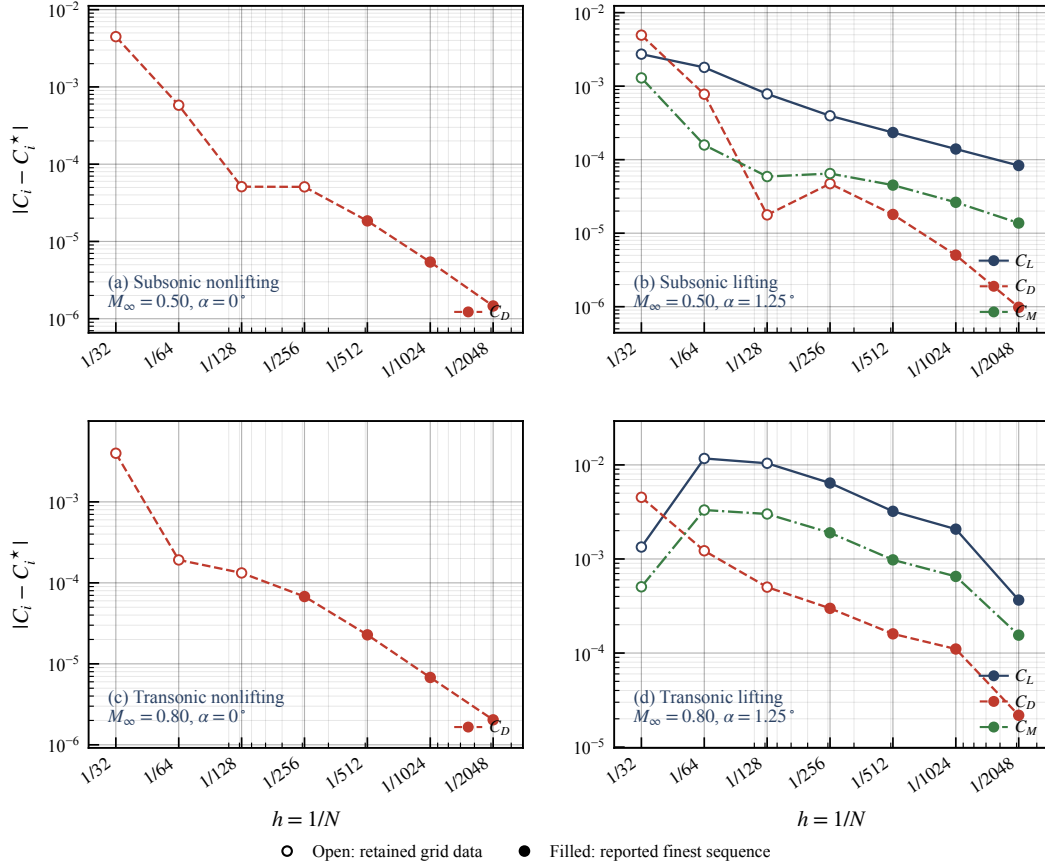


Figure 3.3 Grid-convergence errors  $|C_i - C_i^*|$  versus  $h = 1/N$  for the four Euler benchmark cases. Filled symbols indicate the grid sequence used for the reported positive-order estimates.

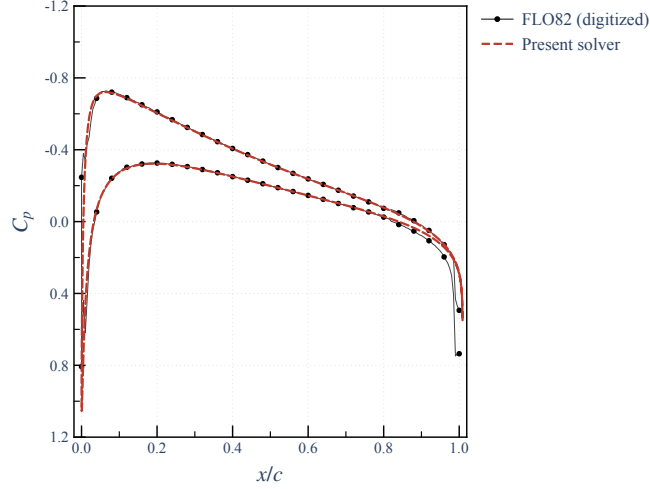


Figure 3.4 Surface-pressure coefficient for the subsonic lifting case,  $M_\infty = 0.50$ ,  $\alpha = 1.25^\circ$ , compared with digitized FLO82 data from Vassberg and Jameson (2010), Fig. 14.

### 3.2 Limitations and Numerical Interpretation

The centred spatial discretization is formally second-order in smooth regions, but that property does not require every aerodynamic functional to display an observed order of exactly two on every finite grid sequence. In the model

$$Q_h = Q^\star + Ch^p,$$

the three-grid estimate is meaningful only when one error term dominates and the sequence is sufficiently close to its asymptotic regime. Before that point, several truncation-error terms may contribute. The integrated quantities  $C_L$ ,  $C_D$ , and  $C_M$  can consequently exhibit different orders, while transonic shocks reduce solution regularity and further modify the observed trend.

This interpretation is consistent with the benchmark study of Vassberg and Jameson (Vassberg and Jameson, 2010). Even for their carefully constructed O-meshes and machine-level residual convergence, the FLO82 subcritical solutions did not reach asymptotic second-order behaviour until very fine grids, and the transonic orders deteriorated. The present orders should therefore be read as finite-sequence indicators, not as universal formal orders. In particular,  $C_L$  and  $C_M$  in the nonlifting cases remain at the numerical symmetry floor and should not be assigned artificial orders. Small grid-to-grid differences can also be contaminated by iterative error, so residual convergence, coefficient stability, grid convergence, and

asymptotic convergence remain distinct criteria.

Figure 3.3 provides the corresponding visual check of this distinction. Several selected refinements are close to linear trends in log-log coordinates, but their slopes vary with the coefficient and flow regime. The finest transonic lifting triplet does not follow a single asymptotic power law, consistently with its negative  $p_{3g}$  values.

All four campaigns reach  $2048^2$ . The subsonic finest-grid coefficients are stable although their residual histories remain finite and oscillatory; the transonic  $2048^2$  runs are fully residual-converged. The transonic lifting case illustrates why agreement with a continuum value and a meaningful Richardson order are separate questions: its direct  $2048^2$  discrepancies are only 0.103%, 0.096%, and 0.400% for  $C_L$ ,  $C_D$ , and  $C_M$ , whereas the (512, 1024, 2048) orders are negative. The successive differences therefore do not support a single dominant error term; more generally, non-monotonic sequences or increasing successive differences invalidate the single-term Richardson model. No Richardson extrapolation is made from that triplet. This negative order indicates a non-asymptotic sequence, not by itself an inaccurate finest-grid solution.

The multigrid result in Fig. 3.1 is a useful iterative acceleration, but it does not replace these verification criteria. The remaining limitations are the finite subsonic residual histories and the absence of a clean single-term asymptotic regime for the finest transonic lifting triplet.

## CHAPTER 4    AEROELASTIC MODELLING AND SOLVER METHODOLOGY

This chapter documents the Michon flap–nonlinear energy sink (flap–NES) model. It separates the experimental configuration and restoring laws published by García Pérez et al. (García Pérez *et al.*, 2025), the laboratory UVLM framework described by Ayala and Parenteau (Ayala, 2025; Parenteau, 2021), and the configuration-specific adaptations introduced here. The latter are implementation choices, not experimental properties; results are reported in Chapter 5.

### 4.1    Aeroelastic Problem and Modelling Strategy

Aeroelasticity couples the structural state  $(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})$  to generalized aerodynamic loads  $\mathbf{Q}_a$ : structural motion changes the flow boundary, and the resulting loads feed back into the structure. The term fluid–structure interaction (FSI) denotes this exchange below. Flutter is used only as the associated dynamic-instability concept; this chapter does not analyse the stability or response of a particular case.

The flap–NES configuration is represented by a low-order structural model with three generalized coordinates. Because the flap is both a structural degree of freedom and an aerodynamic surface, its motion and aerodynamic moment must be exchanged consistently with the UVLM. The FSI exchange is therefore iterated at each new time level.

#### 4.1.1    Selection of the aerodynamic descriptions

Two aerodynamic descriptions address different modelling needs. The Jones realization of Theodorsen’s theory is an inexpensive sectional reference model that represents circulatory memory with a small number of lag states. It is restricted to linear, attached, two-dimensional potential flow and is not used to represent the finite-span flap geometry.

The UVLM represents the three-dimensional, finite-span configuration with a surface discretization, several moving lifting surfaces, and a history of shed circulation (Ayala, 2025; Parenteau, 2021). It assumes potential, inviscid, incompressible, irrotational flow and contains no viscous, separated-flow, stall, or compressibility model; the NACA 0020 section does not alter these assumptions.

The inherited TinyVLM framework provides structured lifting-surface data, the global vortex-lattice system, Biot–Savart kernels, the kinematic-tree interface, and time-marching data (Ayala, 2025; Parenteau, 2021). The partial-span surfaces, structural parameter mapping, flap

kinematics, surface connections, wake grouping, and generalized-load reduction are Michon-specific adaptations; their detailed assembly is given in Appendix B.3.

## 4.2 Michon Experimental Configuration

The published experiment consists of a rigid straight rectangular wing with a NACA 0020 section, supported so that its principal motion is a vertical translation and a rotation about a spanwise elastic axis. The partial-span flap is attached to the aft part of the wing and rotates about a second spanwise hinge. García Pérez et al. report the principal dimensions, the pitch restoring law, and the flap spring data used to define the experimental configurations (García Pérez *et al.*, 2025).

The generalized-coordinate vector used throughout this chapter is

$$\mathbf{q} = \begin{bmatrix} h & \alpha & \beta \end{bmatrix}^T, \quad (4.1)$$

where  $h$  is the plunge displacement,  $\alpha$  is the main-body pitch about the elastic axis, and  $\beta$  is the flap rotation relative to the main body. The sign convention is  $h > 0$  for a translation in the negative global  $z$ -direction. Both angular coordinates are measured in radians and positive rotations use the spanwise  $y$ -axis convention of the kinematic backend.

Table 4.1 Geometric quantities used for the Michon configuration. Values in the first group are reported by García Pérez et al.; the last two quantities are direct geometric conversions used by the present mesh generator.

| Quantity              | Symbol               | Value              | Provenance |
|-----------------------|----------------------|--------------------|------------|
| Airfoil section       | –                    | NACA 0020          | published  |
| Chord                 | $c$                  | 0.200 m            | published  |
| Total span            | $B$                  | 0.440 m            | published  |
| Elastic-axis position | $x_{EA}$             | 0.050 m = 0.25 $c$ | published  |
| Flap-hinge position   | $x_h$                | 0.155 m            | published  |
| Flap chord            | $c_f$                | 0.055 m            | published  |
| Flap span             | $B_f$                | 0.130 m            | published  |
| Flap span fraction    | $B_f/B$              | 0.2955             | conversion |
| Flap leading edge     | $x_{f,LE} = c - c_f$ | 0.145 m            | conversion |

The flap hinge is 45 mm upstream of the main trailing edge. In the partial-span UVLM mesh, the main surface is split into a full-span forward surface, two aft outboard strips, and a central flap surface. This partition preserves the physical span of the flap while allowing the flap surface to have its own kinematic node and force group. It is a surface topology introduced for the present Michon implementation, not a statement that the experimental

wing is physically discontinuous.

Two configurations are defined. In the *OFF* configuration the flap is mechanically locked and

$$\beta = \dot{\beta} = \ddot{\beta} = 0, \quad \mathbf{q}_{\text{act}} = \begin{bmatrix} h & \alpha \end{bmatrix}^T. \quad (4.2)$$

In the *ON* configuration the flap is released and all three coordinates are advanced. The flap is not treated as an independently translating rigid body: its absolute position is obtained by composing its relative rotation with the main-body transformation.

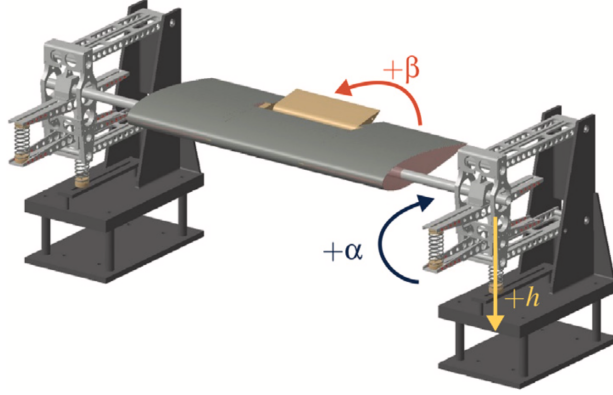


Figure 4.1 Geometry and generalized coordinates of the Michon flap–NES configuration.

#### 4.2.1 Parameter provenance

The published geometry is not sufficient to determine every entry of the structural matrices. The 2.4 kg wing mass and the restoring laws are reported experimentally, whereas some mass moments of the mounted assembly and damping ratios are not reported as complete measured data in the available configuration data. The provenance must therefore be retained with the equations rather than hidden in a single parameter table. The principal values are:

$$M_\alpha(\alpha) = 13.18\alpha + 16.12\alpha^3, \quad M_\beta(\beta) = 1.108\beta + 7.755\beta^3 \quad (4.3)$$

for the pitch and SPF1 flap restoring moments, respectively, with moments in N m and angles in radians (García Pérez *et al.*, 2025). The released configuration also includes the 40 g downstream mass at 80 mm from the main trailing edge. The present ON configuration therefore uses a 2.44 kg normalization mass after adding this point mass to the reported 2.4 kg assembly mass.

The input damping ratios are retained as configuration data, but the available data mark

them as temporary and not identified from the Michon article. Likewise, the flap inertia and static mass moment used by the ON configuration are estimates assembled from the available mass and geometry; their construction is given in Appendix B.1. The inferred-balanced preset is a separate modelling hypothesis and is not used here as a published experimental property.

### 4.3 Three-Degree-of-Freedom Structural Model

#### 4.3.1 Dimensional equations and generalized coordinates

The structural dynamics are written in physical coordinates as

$$\mathbf{M}_{\text{dim}}\ddot{\mathbf{q}} + \mathbf{C}_{\text{dim}}\dot{\mathbf{q}} + \mathbf{K}_{\text{dim}}\mathbf{q} + \mathbf{f}_{\text{nl}}(\mathbf{q}) = \mathbf{Q}_a, \quad (4.4)$$

where  $\mathbf{Q}_a$  contains the aerodynamic force conjugate to  $h$  and the aerodynamic moments conjugate to  $\alpha$  and  $\beta$ . For a released flap, the mass matrix used by the solver is

$$\mathbf{M}_{\text{dim}} = \begin{bmatrix} m_t & S_\alpha & S_\beta \\ S_\alpha & I_\alpha & (x_h - x_{EA})S_\beta + I_\beta \\ S_\beta & (x_h - x_{EA})S_\beta + I_\beta & I_\beta \end{bmatrix}. \quad (4.5)$$

Here  $m_t$  is the translational mass,  $S_\alpha$  is the mass first moment about the elastic axis,  $S_\beta$  is the flap first moment about its hinge, and  $I_\alpha$  and  $I_\beta$  are the corresponding second moments. The off-diagonal term in the  $(\alpha, \beta)$  block follows from expressing the absolute flap velocity as the main-body pitch velocity plus the relative flap velocity. Thus, it is a kinematic consequence of the chosen coordinates, not an additional aerodynamic coupling coefficient.

The linear structural matrices are diagonal in the physical coordinates,

$$\mathbf{K}_{\text{dim}} = \text{diag}(k_h, k_{\alpha 1}, k_{\beta 1}), \quad \mathbf{C}_{\text{dim}} = \text{diag}(2m_t\omega_h\zeta_h, 2I_\alpha\omega_\alpha\zeta_\alpha, 2I_\beta\omega_\beta\zeta_\beta). \quad (4.6)$$

The nonlinear restoring vector is

$$\mathbf{f}_{\text{nl}}(\mathbf{q}) = \begin{bmatrix} 0 \\ k_{\alpha 3}\alpha^3 \\ k_{\beta 3}\beta^3 \end{bmatrix}. \quad (4.7)$$

The dimensional restoring-law mode retains the linear stiffnesses and evaluates the cubic increments within the structural-aerodynamic coupling iteration. When the flap is locked, the physical system is the active  $(h, \alpha)$  subsystem; the fixed-size storage and its reduction

are detailed in Appendix B.1.

### 4.3.2 Nondimensional form

Let  $b = c/2$  be the semi-chord and define

$$\mathbf{x} = \begin{bmatrix} h/b & \alpha & \beta \end{bmatrix}^T, \quad \tau = \omega_\alpha t, \quad V = \frac{U}{b\omega_\alpha}, \quad \mu = \frac{m}{\pi\rho b^2}. \quad (4.8)$$

If  $\mathbf{q} = \mathbf{D}\mathbf{x}$  with  $\mathbf{D} = \text{diag}(b, 1, 1)$ , multiplication of Eq. (4.4) by  $\mathbf{D}^T/(mb^2\omega_\alpha^2)$  gives

$$\mathbf{M}_s \mathbf{x}'' + \mathbf{C}_s \mathbf{x}' + \mathbf{K}_s \mathbf{x} + \mathbf{f}_{\text{nl}}^*(\mathbf{x}) = \mathbf{Q}_a^*, \quad (4.9)$$

where primes denote derivatives with respect to  $\tau$ . The normalized force scales are

$$Q_h^* = \frac{Q_h}{mb\omega_\alpha^2}, \quad Q_\alpha^* = \frac{Q_\alpha}{mb^2\omega_\alpha^2}, \quad Q_\beta^* = \frac{Q_\beta}{mb^2\omega_\alpha^2}. \quad (4.10)$$

Here  $a = x_{EA}/b - 1$  and  $c_h = x_h/b - 1$  are the dimensionless elastic-axis and hinge locations measured from mid-chord. The normalized matrices and mass properties are collected in Appendix B.1.

## 4.4 Unsteady Aerodynamic Models

### 4.4.1 Jones realization of Theodorsen aerodynamics

Theodorsen's theory gives the circulatory response of a two-dimensional thin airfoil undergoing harmonic motion under linear potential-flow assumptions (Theodorsen, 1949). Its frequency-domain function is useful as a compact reference, but direct frequency-domain evaluation is not a time-marching state equation. The present reduced model therefore uses the Jones approximation of Wagner's indicial function (Jones, 1938):

$$\Phi(s) \simeq 1 - 0.165 \exp(-0.0455s) - 0.335 \exp(-0.300s), \quad s = \frac{Ut}{b}. \quad (4.11)$$

The two exponential terms are represented by lag states. For a scalar downwash quantity  $w$ , a generic lag state satisfies

$$\frac{d\lambda_j}{ds} = b_j \ell_j w - \ell_j \lambda_j, \quad (b_1, b_2) = (-0.165, -0.335), \quad (\ell_1, \ell_2) = (0.0455, 0.300). \quad (4.12)$$

The three-coordinate implementation applies the same realization to the generalized downwash generated by plunge, pitch, and flap motion. After collecting the added-mass, damping,

stiffness, and lag contributions, its state-space form is

$$\mathbf{Q}_a = \mathbf{M}_a \mathbf{x}'' + \mathbf{D}_a \mathbf{x}' + \mathbf{K}_a \mathbf{x} + \mathbf{L}_\delta \boldsymbol{\lambda}, \quad \boldsymbol{\lambda}' = \mathbf{Q}_{\lambda a} \mathbf{x}'' + \mathbf{Q}_{\lambda v} \mathbf{x}' + \mathbf{L}_\lambda \boldsymbol{\lambda}. \quad (4.13)$$

Substitution in Eq. (4.9) gives the acceleration equation solved by the reduced path:

$$(\mathbf{M}_s - \mathbf{M}_a) \mathbf{x}'' = (\mathbf{D}_a - \mathbf{C}_s) \mathbf{x}' + (\mathbf{K}_a - \mathbf{K}_s) \mathbf{x} + \mathbf{L}_\delta \boldsymbol{\lambda} - \mathbf{f}_{\text{nl}}^*(\mathbf{x}). \quad (4.14)$$

The detailed geometry coefficients  $T_i(a, c_h)$  and their matrix assembly are given in Appendix B.2; they are algebraic expansion terms of the same reduced model, not additional physical assumptions. This model is a baseline for a low-order, attached-flow description; it does not contain a finite-span wake or a separated-flow correction.

Both time-domain implementations represent the same restoring laws, with different allocations between the linear matrices and nonlinear increment. The reduced model retains the linear pitch stiffness and supplies the flap law through its nonlinear term; the UVLM retains the linear flap stiffness and subtracts only the cubic increment in Eq. (4.31). These are implementation choices, not different physical configurations.

For time integration, the reduced Jones model is written as a first-order system. With the mathematical ordering

$$\mathbf{y} = [\mathbf{x} \quad \mathbf{x}' \quad \boldsymbol{\lambda}]^T, \quad \mathbf{y}' = \begin{bmatrix} \mathbf{x}' \\ \mathbf{a}(\mathbf{x}, \mathbf{x}', \boldsymbol{\lambda}) \\ \mathbf{Q}_{\lambda a} \mathbf{a} + \mathbf{Q}_{\lambda v} \mathbf{x}' + \mathbf{L}_\lambda \boldsymbol{\lambda} \end{bmatrix}, \quad (4.15)$$

where  $\mathbf{a}$  is obtained from Eq. (4.14). State-ordering and matrix-assembly details are given in Appendix B.2; the system is integrated with an adaptive RK45 method.

#### 4.4.2 Multi-surface UVLM formulation

For the three-dimensional path, the flow is represented by a potential  $\phi$  satisfying

$$\mathbf{u} = \nabla \phi, \quad \nabla^2 \phi = 0, \quad (4.16)$$

away from the vortex filaments. Each configured lifting surface is discretized into structured quadrilateral panels carrying bound vortex rings. The circulation unknowns are obtained by imposing no penetration at each panel collocation point:

$$\mathbf{n}_i^T (\mathbf{V}_{\text{body},i} + \mathbf{V}_{\text{bound},i}(\boldsymbol{\Gamma}_b) + \mathbf{V}_{\text{wake},i}(\boldsymbol{\Gamma}_w)) = 0, \quad \mathbf{A}(\mathbf{q}) \boldsymbol{\Gamma}_b = \mathbf{b}(\mathbf{q}, \dot{\mathbf{q}}, \boldsymbol{\Gamma}_w). \quad (4.17)$$

The influence coefficients are evaluated from finite vortex-segment Biot–Savart kernels with the regularization used by the backend. The matrix is global over all lifting surfaces, so main–flap mutual induction is included in the same circulation solve. For the Michon configuration, the bound lattice comprises four surfaces: a full-span forward surface, two aft outboard surfaces, and a central flap surface; this topology is shown in Fig. 4.2.

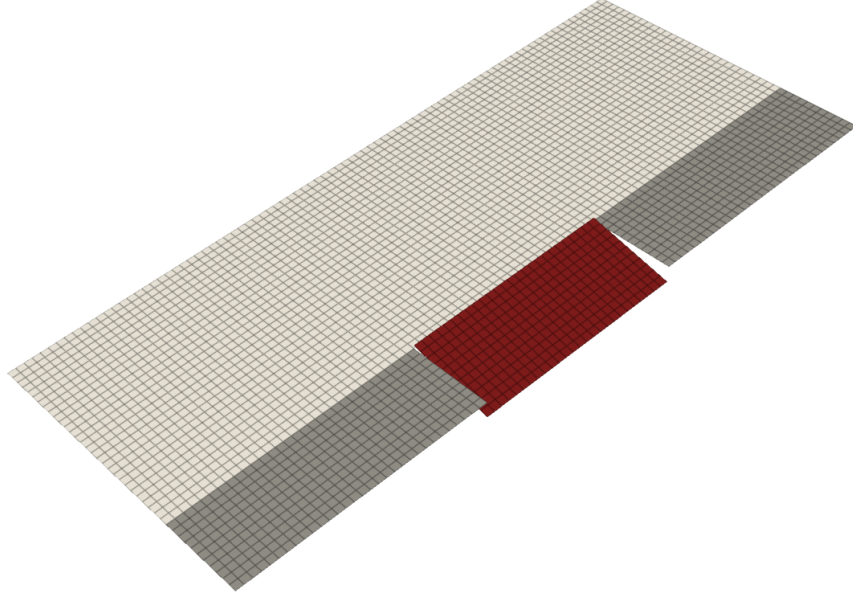


Figure 4.2 Michon UVLM multi-surface discretization in the undeformed configuration. The bound lattice comprises the full-span forward surface, two aft outboard surfaces, and the central flap surface.

The present UVLM path separates three data sets: bound circulation  $\mathbf{\Gamma}_b$ , wake circulation  $\mathbf{\Gamma}_w$ , and wake vertices  $\mathbf{X}_w$ . A new wake station is copied from the current trailing edge at each time-level update. During a fluid–structure subiteration, the already accepted wake circulation and geometry are used in the wake-influence right-hand side; the candidate trailing-edge circulation is not committed as an accepted wake row until the coupled step is accepted. The default Michon path therefore uses a frozen, history-based wake in the influence calculation and does not execute an independent free-wake convection or roll-up operation.

#### 4.4.3 Michon multi-surface kinematics

The kinematic tree is the main adaptation required by the partial-span flap. The main-body node is a translation and a rotation about the elastic axis; the flap node is a relative rotation about the hinge and is attached as a child of the main node. In the local evaluation time,

the transforms can be written as

$$\mathbf{T}_m(t; \tau_{n+1}) = \mathbf{T}_z(-h - \dot{h}(t - \tau_{n+1})) \mathbf{R}_y(x_{EA}, \alpha + \dot{\alpha}(t - \tau_{n+1})), \quad (4.18)$$

$$\mathbf{T}_f(t; \tau_{n+1}) = \mathbf{R}_y(x_h, \beta + \dot{\beta}(t - \tau_{n+1})), \quad (4.19)$$

$$\mathbf{T}_{f,\text{global}} = \mathbf{T}_m \mathbf{T}_f. \quad (4.20)$$

The formulation evaluates the transforms and their time derivatives with dual numbers, so panel positions and boundary-condition velocities come from the same transformation rather than separate finite differences. When  $\beta$  is locked, the flap node remains attached to the main-body configuration with zero relative state; further assembly details are given in Appendix B.3.1.

The surface partition and parent–child graph are Michon-specific adaptations of the inherited kinematic-tree framework.

#### 4.5 Aerodynamic Forces and Generalized Loads

The circulation distribution is first converted into chordwise circulation jumps. For a panel with chordwise index  $j$  and spanwise index  $i$ , the implementation uses

$$\Delta\Gamma_{i,0} = \Gamma_{i,0} - \Gamma_{\text{upstream TE},i}, \quad \Delta\Gamma_{i,j} = \Gamma_{i,j} - \Gamma_{i,j-1}, \quad j > 0, \quad (4.21)$$

where the upstream term is present only when a chordwise surface connection exists. This makes the circulation difference continuous across the main surface/flap interface as represented by the configured topology.

For each panel, the CPU force kernel computes the working-density form of the Kutta–Joukowski and unsteady circulation terms,

$$\mathbf{f}_i^{\text{circ}} = \Delta\Gamma_i (\mathbf{V}_i \times \Delta\mathbf{y}_i), \quad \mathbf{f}_i^{\text{unsteady}} = \dot{\Gamma}_i A_i \mathbf{n}_i, \quad \mathbf{f}_i = \mathbf{f}_i^{\text{circ}} + \mathbf{f}_i^{\text{unsteady}}. \quad (4.22)$$

The backend stores the working-density contribution with  $\rho = 1$ ; the physical density is introduced when the loads are reduced to the normalized structural equations. This implementation detail is why the density should not be inserted a second time into Eq. (4.22). The force application point is the midpoint of the spanwise bound-vortex segment associated with the panel.

The total panel force is reduced about the transformed elastic-axis point for pitch and about the transformed flap-hinge point for flap rotation. With  $\eta = \rho b^3/m$ , the normalized general-

ized loads are

$$Q_h^* = -\eta \sum_i f_{i,z}, \quad (4.23)$$

$$Q_\alpha^* = \eta \sum_i (\mathbf{r}_{EA,i} \times \mathbf{f}_i)_y, \quad (4.24)$$

$$Q_\beta^* = \eta \sum_{i \in \text{flap}} (\mathbf{r}_{h,i} \times \mathbf{f}_i)_y. \quad (4.25)$$

The last component is set to zero in the locked configuration. The circulatory and unsteady components are reduced separately before their sum is transferred to the structural solver. Dimensional reconstruction uses  $mb\omega_\alpha^2$  for the plunge-force scale and  $mb^2\omega_\alpha^2$  for the two moment scales.

## 4.6 Time Integration and Implicit Fluid–Structure Coupling

### 4.6.1 Newmark structural update

The structural equation is advanced with the Newmark family using

$$\beta_N = \frac{1}{4}, \quad \gamma_N = \frac{1}{2}. \quad (4.26)$$

For a linear structural system, these values define the average-acceleration scheme. Under the standard assumptions of linear structural dynamics, the scheme is second-order accurate and unconditionally stable; this statement does not extend automatically to the nonlinear, partitioned fluid–structure iteration (Newmark, 1959; Ayala, 2025).

For a normalized time step  $\Delta\tau$ , the Newmark relations are

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta\tau \mathbf{x}'_n + \Delta\tau^2 \left[ \left( \frac{1}{2} - \beta_N \right) \mathbf{x}''_n + \beta_N \mathbf{x}''_{n+1} \right], \quad (4.27)$$

$$\mathbf{x}'_{n+1} = \mathbf{x}'_n + \Delta\tau \left[ (1 - \gamma_N) \mathbf{x}''_n + \gamma_N \mathbf{x}''_{n+1} \right]. \quad (4.28)$$

Eliminating the new acceleration gives the effective operator

$$\mathbf{K}_{\text{eff}} = \mathbf{K}_s + a_0 \mathbf{M}_s + a_1 \mathbf{C}_s, \quad a_0 = \frac{1}{\beta_N \Delta\tau^2}, \quad a_1 = \frac{\gamma_N}{\beta_N \Delta\tau}. \quad (4.29)$$

The integrator solves for an increment  $\Delta\mathbf{x}$  with

$$\mathbf{K}_{\text{eff}} \Delta\mathbf{x} = \Delta\mathbf{F} + \mathbf{M}_s (x_{d0} \mathbf{x}'_n + x_{dd0} \mathbf{x}''_n) + \mathbf{C}_s (x_{d1} \mathbf{x}'_n + x_{dd1} \mathbf{x}''_n), \quad (4.30)$$

where the coefficient definitions and their relation to the classical Newmark form are given in Appendix B.4.

The aerodynamic-force increment used in the present dimensional-restoring-law path is

$$\Delta \mathbf{F} = \mathbf{Q}_a^{n+1,k} - \mathbf{Q}_a^n - \left[ \mathbf{f}_{\text{nl}}^*(\mathbf{x}^{n+1,k}) - \mathbf{f}_{\text{nl}}^*(\mathbf{x}^n) \right]. \quad (4.31)$$

Thus, the cubic restoring-force change is not hidden inside the linear effective matrix.

The transient UVLM step is chosen from the trailing-edge panel chord and the root convective velocity,

$$\Delta \tau_{\text{conv}} = \frac{\ell_{\text{TE}}}{\|\mathbf{V}_{\text{root}}\|}, \quad \Delta \tau = \min(\Delta \tau_{\text{conv}}, \Delta \tau_{\text{max}}) \quad \text{when a user cap is specified.} \quad (4.32)$$

This keeps the temporal spacing tied to the spatial wake spacing used by the time-marching UVLM.

#### 4.6.2 Partitioned implicit algorithm

The aerodynamic and structural operators are kept separate, but the exchange at a physical time step is iterated to obtain a consistent state and load at the same time level (Ayala, 2025). The coupling sequence is

---

**Algorithm 1** Implicit UVLM–Structural Coupling

---

```

1:  $\mathcal{S}^n \leftarrow (\mathbf{q}^n, \dot{\mathbf{q}}^n, \ddot{\mathbf{q}}^n, \mathbf{\Gamma}_b^n, \mathbf{X}_w^n, \mathbf{\Gamma}_w^n)$ 
2:  $\Delta \mathbf{F}^{(0)} \leftarrow \mathbf{0}$ 
3: repeat
4:    $\Delta \mathbf{F}^{(k)} \xrightarrow{\mathcal{N}_{\Delta\tau}} (\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})^{(k+1)}$ 
5:   Evaluate structural residuals  $(r_q, r_{\dot{q}}, r_{\ddot{q}})$  on the active degrees of freedom
6:   Apply the configured relaxation to the structural increments ▷ Relaxation
7:   Update candidate geometry and panel kinematics
8:    $(\mathbf{X}_{TE}^{(k+1)}, \mathbf{X}_w^n) \rightarrow \widetilde{\mathbf{X}}_w^{(k+1)}$  ▷ Accepted wake history is fixed
9:    $\mathbf{A}^{(k+1)} \mathbf{\Gamma}_b^{(k+1)} = \mathbf{b}^{(k+1)}(\mathbf{\Gamma}_w^n)$ 
10:   $\mathbf{\Gamma}_b^{(k+1)} \rightarrow (\Delta \mathbf{\Gamma}, \dot{\mathbf{\Gamma}}) \rightarrow \mathbf{Q}_a^{(k+1)}$ 
11:  Form  $\Delta \mathbf{F}^{(k+1)}$  from  $\mathbf{Q}_a^{(k+1)}$  and the nonlinear restoring-force increment
12:  Evaluate  $(r_q, r_{\dot{q}}, r_{\ddot{q}}, r_Q)$  and the convergence gate
13:   $k \leftarrow k + 1$ 
14: until  $\mathcal{G}(r_q, r_{\dot{q}}, r_{\ddot{q}}, r_Q)$  is satisfied
15:  $\mathbf{X}_w^{n+1} \leftarrow \widetilde{\mathbf{X}}_w^{(k)}$ ,  $\mathbf{\Gamma}_w^{n+1} \leftarrow \text{append}(\mathbf{\Gamma}_w^n, \mathbf{\Gamma}_{TE}^{(k)})$  ▷ Commit wake after convergence
16:  $\mathcal{S}^{n+1} \leftarrow (\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \mathbf{\Gamma}_b)^{(k)}$  ▷ Commit state and bound circulation

```

---

The coupling is therefore implicit and iterated at each physical time step. The gate uses separate state and load residuals; the accepted wake history remains fixed during subiterations, and the new wake row is committed only after convergence. Complete residual definitions, relaxation, and gate details are given in Appendix B.4. Newmark stability alone does not guarantee stability of the complete nonlinear FSI iteration.

## CHAPTER 5 AEROELASTIC NUMERICAL RESULTS

### 5.1 Aerodynamic Verification and Reference-Model Validation

The aerodynamic implementation is assessed against analytical reference problems of increasing complexity: thin-airfoil circulation, articulated-flap response, and prescribed harmonic/biharmonic motion. Comparisons with the experimental data reported by García Pérez et al. are considered separately in Section 5.2.

#### 5.1.1 Quasi-two-dimensional steady benchmarks

##### Thin-airfoil Fourier coefficients

The first test projects the circulation exported by the multi-surface UVLM onto the first two thin-airfoil Fourier coefficients. The static cases use the Michon geometry in a quasi-two-dimensional limit with  $c = 0.2$  m,  $AR = 256$ ,  $U = 16$  m s<sup>-1</sup>, and  $\beta = 0$ . The prescribed pitch angles are  $\alpha \in \{-4, -2, 0, 2, 4\}^\circ$ . The projection is performed on the integrated circulation jump of each physical chord segment, rather than on a pointwise leading-edge circulation density. With  $\theta = \arccos(1 - 2x/c)$ , the fitted segment circulation is

$$\Delta\Gamma_j = UcA_0 [\theta + \sin\theta]_{\theta_L}^{\theta_R} + \frac{Uc}{2}A_1 \left[ \theta - \frac{1}{2}\sin(2\theta) \right]_{\theta_L}^{\theta_R}, \quad (5.1)$$

and the corresponding lift coefficient is reconstructed as

$$C_l^{A_0A_1} = 2\pi \left( A_0 + \frac{A_1}{2} \right). \quad (5.2)$$

This test therefore checks both the Fourier projection of the exported circulation jumps and the consistency of their integrated lift reconstruction. The exported static system contains 4464 panels and uses spanwise mirror symmetry for this test.

The largest deviation between  $A_0$  and the imposed angle is  $5.83 \times 10^{-3}$ , while the RMS value of the fitted  $A_1$  coefficient is  $5.89 \times 10^{-3}$ . The maximum difference between the reconstructed and direct UVLM lift coefficients is  $2.35 \times 10^{-3}$ ; the normalized RMS of the direct lift relative to the  $2\pi\alpha$  reference is 1.84%. These values quantify the remaining discrepancy in this quasi-two-dimensional test without treating  $A_0$  or  $A_1$  as solver state variables.

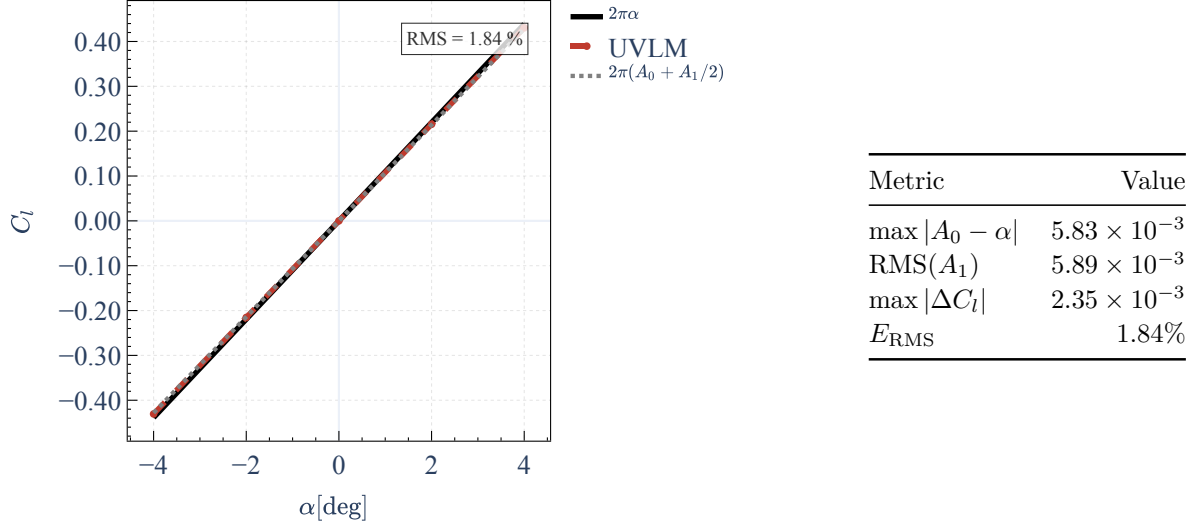


Figure 5.1 Static  $A_0$ – $A_1$  reconstruction for the quasi-two-dimensional Michon geometry: lift comparison (left) and associated reconstruction errors (right). Here  $\Delta C_l = C_l^{A_0 A_1} - C_{l,\text{UVLM}}$ .

### Thin-airfoil flap benchmark

The second static test retains the same  $AR = 256$  quasi-two-dimensional representation and introduces the articulated flap. The hinge is located at  $x_h/c = 0.775$ ; the analysis section is the connected section at  $y = 0$ . With  $\alpha = 0$  and  $\beta \in \{-4, -2, 0, 2, 4\}^\circ$ , the reference coefficients used by the test are

$$\theta_h = \arccos(1 - 2x_h/c), \quad (5.3)$$

$$A_0 = \alpha + \beta \frac{\pi - \theta_h}{\pi}, \quad A_1 = \frac{2\beta \sin \theta_h}{\pi}. \quad (5.4)$$

with  $C_l = 2\pi(A_0 + A_1/2)$ . Positive  $\beta$  is a rotation about  $+\mathbf{e}_y$ , giving a flap camber-line slope  $dz_c/dx = -\beta$  downstream of the hinge. The corresponding full-span static system contains 8928 panels.

The sectional and global slopes are 3.62896 and 3.59215; relative to the thin-airfoil reference slope 3.64719, their differences are 0.50% and 1.51%, respectively. The normalized RMS errors are 2.60% and 2.92%, so the articulated-flap response remains within 3% RMS of the reference.

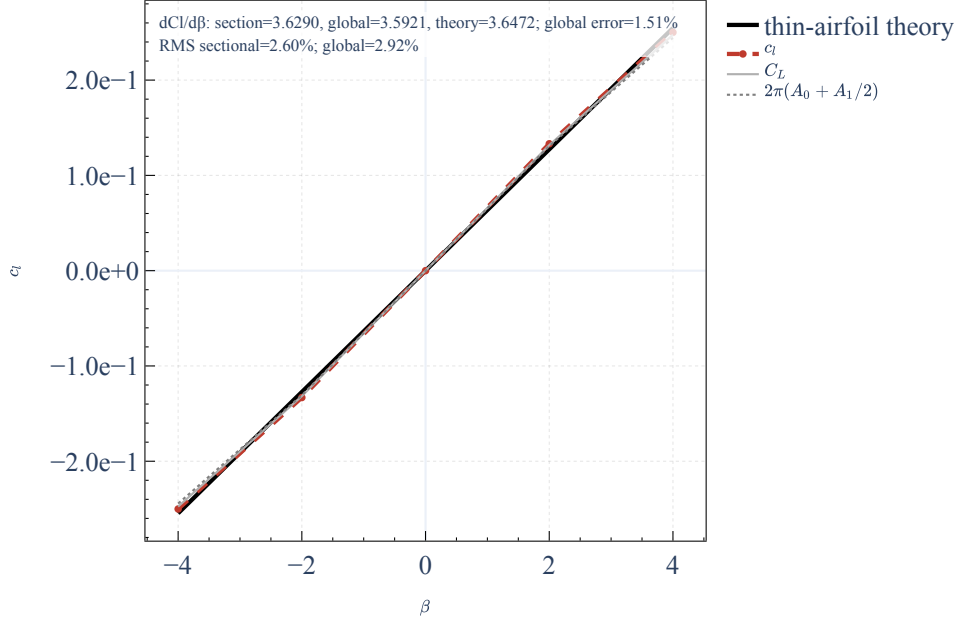


Figure 5.2 Static flap-effect benchmark: sectional and global UVLM lift responses compared with the thin-airfoil flap reference.

### 5.1.2 Harmonic unsteady response

The prescribed-pitch tests use the same quasi-two-dimensional multi-surface geometry,  $AR = 256$ ,  $c = 0.2$  m,  $b = 0.1$  m, and  $U = 16$  m s<sup>-1</sup>. The imposed motion is

$$\alpha(t) = A \sin(\omega t), \quad A = 1^\circ, \quad k = \frac{\omega b}{U}, \quad f = \frac{\omega}{2\pi} = \frac{kU}{2\pi b}. \quad (5.5)$$

The four cases use  $k \in \{0.25, 0.5, 1, 2\}$ ,  $N_c = 40$ , and 80 spanwise panels on the represented half-span. The reference curves are the time-domain Jones realization used for the Theodorsen/Wagner response, rather than a separate exact frequency-domain evaluation.

For each case, the reported error is the relative RMS over the plotted final cycle,

$$E_{\text{RMS}} = 100 \frac{\sqrt{N^{-1} \sum_{i=1}^N (C_{l,i}^{\text{UVLM}} - C_{l,i}^{\text{ref}})^2}}{\max_i |C_{l,i}^{\text{ref}}|}. \quad (5.6)$$

The relative RMS increases from 1.45% at  $k = 0.25$  to 4.65% at  $k = 2$ ; the trend checks the unsteady circulation path and time-domain reference reconstruction but supports no experimental or coupled-structure conclusion.

Table 5.1 Harmonic prescribed-pitch cases and relative RMS errors.

| $k$  | $f$ [Hz] | $N_c$ | $N_{s,1/2}$ | $E_{\text{RMS}}$ [%] |
|------|----------|-------|-------------|----------------------|
| 0.25 | 6.366    | 40    | 80          | 1.45                 |
| 0.50 | 12.732   | 40    | 80          | 2.07                 |
| 1.00 | 25.465   | 40    | 80          | 2.96                 |
| 2.00 | 50.930   | 40    | 80          | 4.65                 |

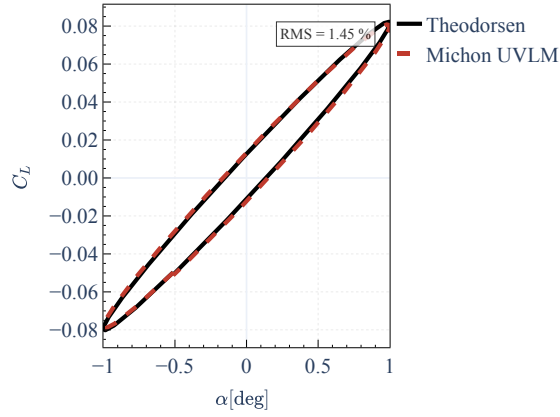
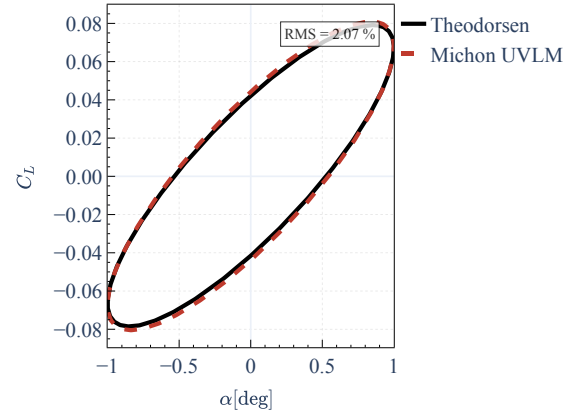
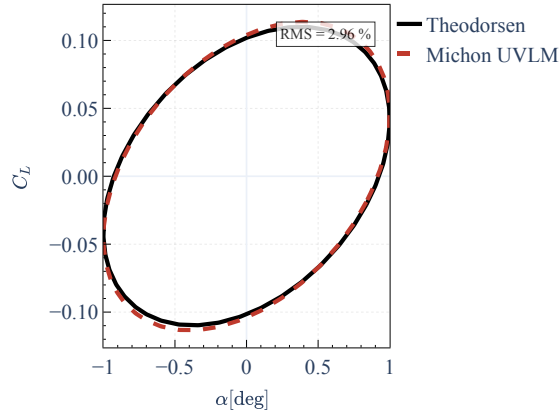
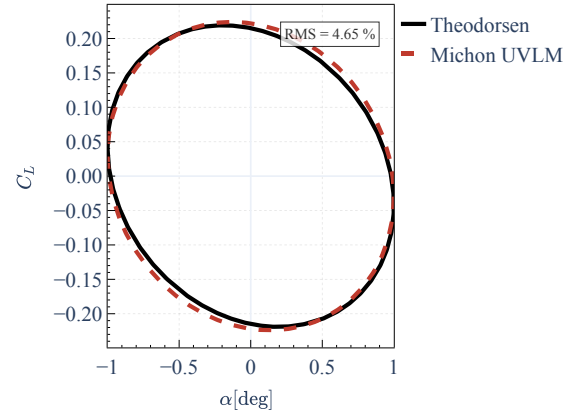
(a)  $k = 0.25$ (b)  $k = 0.5$ (c)  $k = 1.0$ (d)  $k = 2.0$ 

Figure 5.3 Harmonic prescribed-pitch response and Jones realization of the Theodorsen reference.

### 5.1.3 Biharmonic unsteady response

The final reference-model test uses the prescribed motion

$$\alpha(t) = A [\sin(\omega t) + 2 \sin(3\omega t)], \quad A = 1^\circ, \quad k = \frac{\omega b}{U}. \quad (5.7)$$

The two components have reduced frequencies  $k$  and  $3k$ . The cases use  $k \in \{0.25, 0.54, 1, 2\}$ ,  $AR = 256$ ,  $U = 16 \text{ m s}^{-1}$ , and 80 half-span panels; the chordwise resolutions are  $N_c = 40, 80, 80, 80$ , respectively. The value  $k = 0.54$  is retained as specified for this case and is not a relabelling of  $k = 0.5$ .

The total relative RMS uses Eq. (5.6). To quantify the two frequency components separately, the final fundamental-period signals were fitted with complex first- and third-harmonic coefficients. For  $j \in \{1, 3\}$ , the reported component error is

$$E_{Hj} = 100 \frac{|\hat{C}_{l,j}^{\text{UVLM}} - \hat{C}_{l,j}^{\text{ref}}|}{|\hat{C}_{l,j}^{\text{ref}}|}. \quad (5.8)$$

Table 5.2 Biharmonic prescribed-pitch cases and component-wise errors.

| $k$  | $3k$ | $f_1$ [Hz] | $N_c$ | $E_{\text{RMS}}$ [%] | $E_{H1}$ [%] | $E_{H3}$ [%] |
|------|------|------------|-------|----------------------|--------------|--------------|
| 0.25 | 0.75 | 6.366      | 40    | 2.06                 | 2.01         | 3.65         |
| 0.54 | 1.62 | 13.751     | 80    | 2.20                 | 2.98         | 3.48         |
| 1.00 | 3.00 | 25.465     | 80    | 3.30                 | 4.06         | 5.22         |
| 2.00 | 6.00 | 50.930     | 80    | 6.14                 | 6.80         | 9.51         |

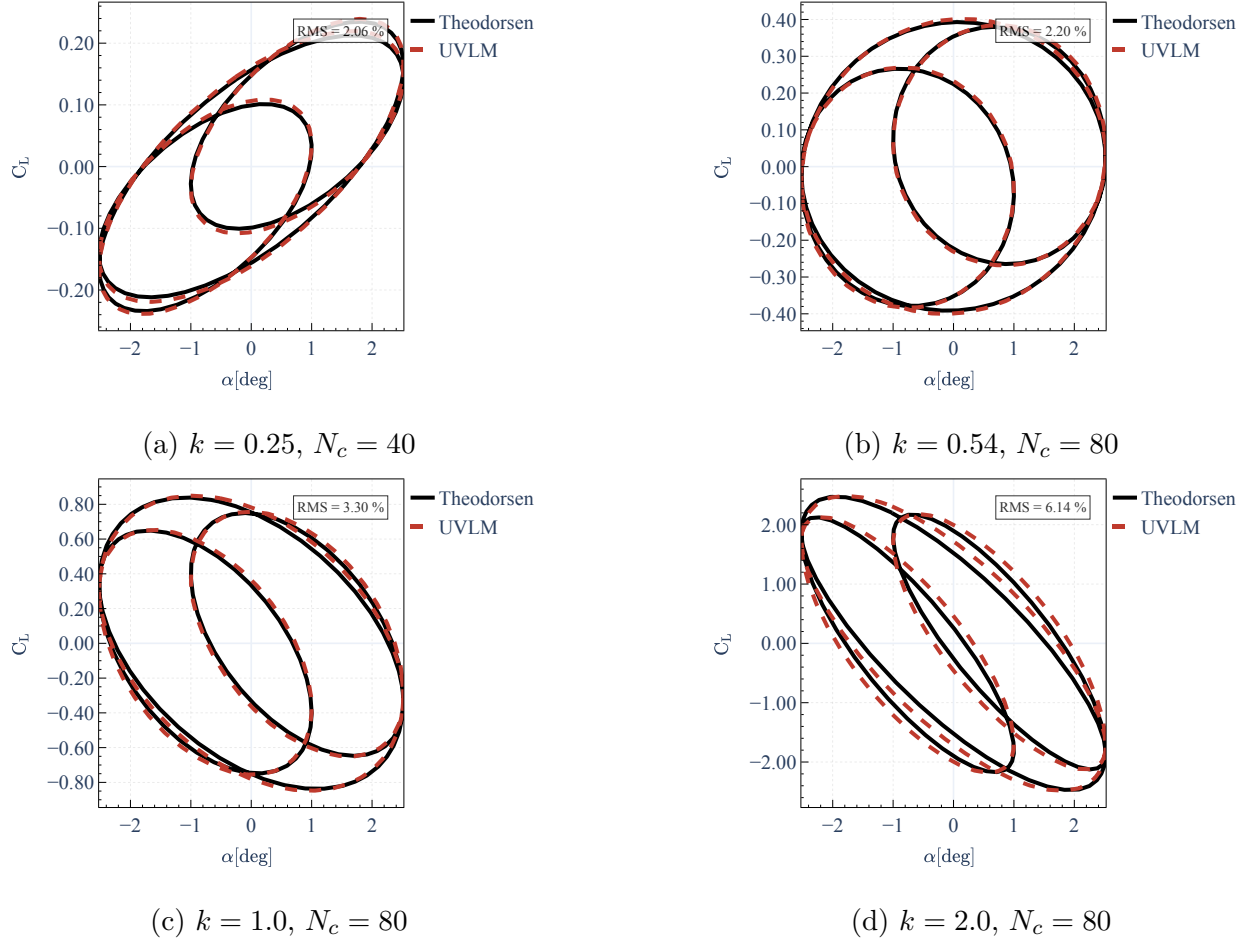


Figure 5.4 Biharmonic prescribed-pitch response and Jones realization of the Theodorsen reference.

The total RMS rises from 2.06% at  $k = 0.25$  to 6.14% at  $k = 2$ , with the fundamental and third-harmonic errors showing the same frequency-dependent trend in Table 5.2. This superposed-excitation test checks the corresponding linear reference reconstruction, not non-linear aerodynamic separation or stall.

## 5.2 Application-Level Comparison with García Pérez et al.’s Measurements

The reconstructed model is compared with the measurements reported by García Pérez et al. at three levels: zero-flow structural frequencies, free-return branches (Fig. 7), and finite-amplitude responses (Fig. 8). Since Figs. 7–8 provide graphical rather than tabulated values, readings from them are approximate and are not treated as raw data (García Pérez *et al.*, 2025). Flutter denotes loss of stability of the coupled equilibrium, whereas a limit-cycle oscillation (LCO) is a finite-amplitude periodic response; the comparisons therefore test the scope of the potential-flow UVLM against the measurements.

### 5.2.1 Zero-flow structural response

At  $U = 0$ , the structural frequencies are obtained from

$$\mathbf{K}\boldsymbol{\phi}_j = \omega_j^2 \mathbf{M}\boldsymbol{\phi}_j, \quad f_j = \frac{\omega_j}{2\pi}. \quad (5.9)$$

Table 5.3 uses the “inferred-balanced” reconstruction: it retains the reported 40 g mass and SPF1 restoring law, but sets  $S_{\alpha,\text{base}} = 0$  and  $I_{\alpha,\text{base}} = 0.014 \text{ kg m}^2$  as inverse hypotheses. The flap mass properties are likewise estimated from the available geometry and point mass. This is a conditional structural check, not an independent identification of the experimental mass matrix.

Table 5.3 Zero-flow frequencies of the inferred-balanced structural reconstruction compared with approximate readings of Michon Fig. 7.

| Configuration | Mode | Dominant coordinate | $f_{\text{calc}}$ [Hz] | $f_{\text{exp}}$ [Hz] | Relative error [%] |
|---------------|------|---------------------|------------------------|-----------------------|--------------------|
| OFF           | 1    | $\alpha$            | 4.5449                 | 4.55                  | 0.11               |
| OFF           | 2    | $h$                 | 6.0171                 | 6.00                  | 0.28               |
| ON            | 1    | $\alpha$            | 4.1596                 | 3.95                  | 5.31               |
| ON            | 2    | $h$                 | 5.9191                 | 5.80                  | 2.05               |
| ON            | 3    | $\beta$             | 7.9446                 | 9.00                  | 11.73              |

The OFF frequencies are reproduced within the accuracy of the graphical readings. In the ON case, the first two modes remain reasonably close, whereas the flap-dominated mode is lower by 1.06 Hz; this discrepancy is compatible with uncertainty in the unreported moving-flap inertia, including the rod, shaft, encoder rotor, and moving spring hardware. The values describe the inferred-balanced reconstruction, not an independent mass-matrix identification.

### 5.2.2 Linear aeroelastic response: Fig. 7

García Pérez et al. identified resonant frequencies from free recoveries at fixed airspeeds and report means and standard deviations from five measurements at each speed (García Pérez *et al.*, 2025). The present calculations use the four-surface Michon geometry, the inferred-balanced structural input, a fine mesh, five-second returns, and small initial perturbations

$$h(0) = 10^{-4} \text{ m}, \quad \alpha(0) = \beta(0) = 0.02^\circ \quad (5.10)$$

for ON;  $\beta(0) = 0$  for OFF. The sampled speeds are  $U \in \{2, 8, 12, 16\} \text{ m s}^{-1}$ . The ON case at  $U = 12 \text{ m s}^{-1}$  does not pass the coupling convergence gate and is excluded from the frequency extraction.

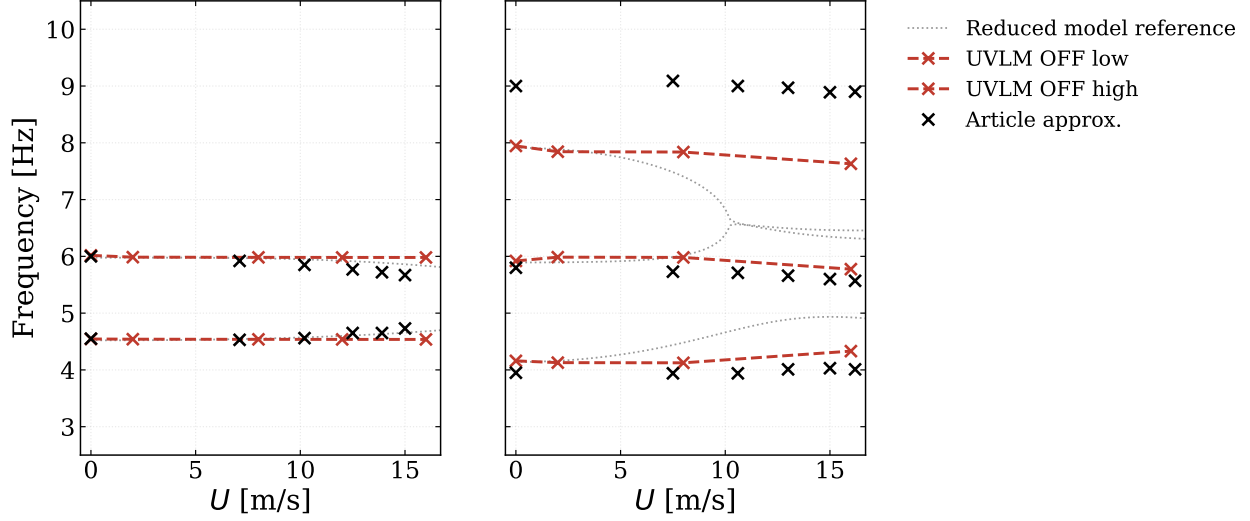


Figure 5.5 Free-return frequencies of the inferred-balanced multi-surface UVLM compared with approximate readings of Fig. 7 of García Pérez et al. The missing ON point at  $U = 12 \text{ m s}^{-1}$  corresponds to a non-converged FSI calculation.

The extracted frequencies are listed in Table 5.4. For the completed OFF cases,

$$|\Delta f_{\text{OFF}}| < 6 \times 10^{-3} \text{ Hz}$$

between  $U = 2$  and  $16 \text{ m s}^{-1}$ , whereas the experimental branches exhibit a marked velocity dependence. The present UVLM therefore does not reproduce the measured Fig. 7 branch evolution. Completed ON cases retain three identifiable numerical peaks, but the flap branch is systematically about 1.1–1.3 Hz below the graphical readings; the ON case at  $U = 12 \text{ m s}^{-1}$  is unavailable because it did not converge.

Table 5.4 UVLM free-return frequencies for the Fig. 7 comparison.

| Configuration | $U$ [m/s] | $f_1$ [Hz] | $f_2$ [Hz] | $f_3$ [Hz] |
|---------------|-----------|------------|------------|------------|
| OFF           | 2         | 4.5409     | 5.9858     | –          |
| OFF           | 8         | 4.5384     | 5.9824     | –          |
| OFF           | 12        | 4.5373     | 5.9810     | –          |
| OFF           | 16        | 4.5371     | 5.9807     | –          |
| ON            | 2         | 4.1281     | 5.9858     | 7.8434     |
| ON            | 8         | 4.1249     | 5.9810     | 7.8372     |
| ON            | 12        | –          | –          | –          |
| ON            | 16        | 4.3308     | 5.7745     | 7.6305     |

### 5.2.3 Nonlinear response: Fig. 8

The experimental Fig. 8 amplitude is the mean of local-extremum amplitudes measured from the zero reference after 30 s of established LCO response; the first speed at which a stable LCO is observed defines the experimental onset (García Pérez *et al.*, 2025). Approximate graphical readings give a last no-LCO point near  $15.30 \text{ m s}^{-1}$  and a first small OFF LCO near  $15.65 \text{ m s}^{-1}$ , compared with approximately  $17.15$  and  $17.40 \text{ m s}^{-1}$  for ON. These values are used only as reference ranges because the raw experimental time histories are unavailable.

The numerical calculations use five-second histories and an analysis window of approximately 2.5 s, with small to large initial perturbations at selected speeds between 12 and  $20 \text{ m s}^{-1}$ . All completed cases are damped; the large-perturbation ON case at  $20 \text{ m s}^{-1}$  is not converged.

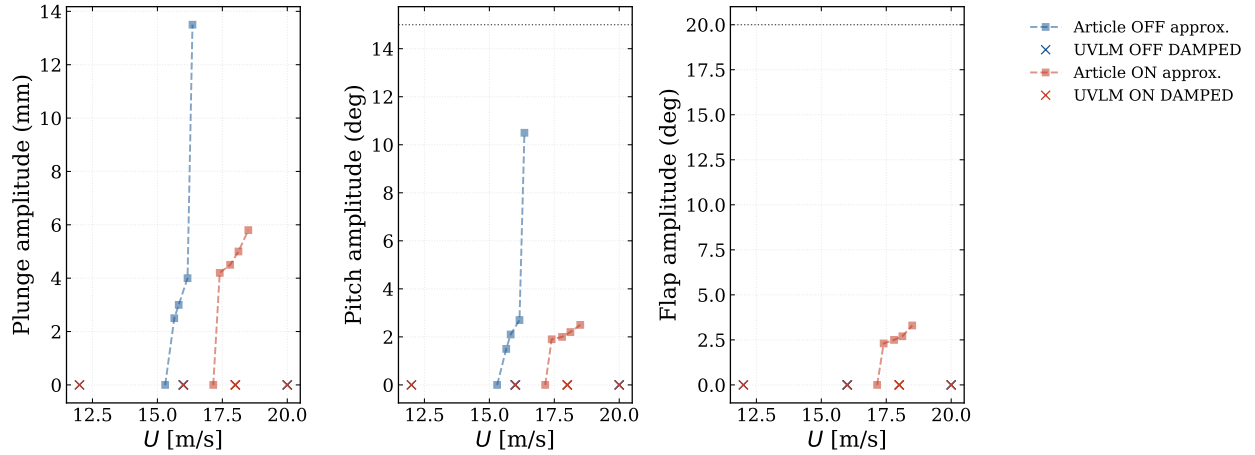


Figure 5.6 Multi-surface UVLM response compared with approximate Fig. 8 amplitude data reported by García Pérez *et al.* Completed numerical cases remain damped; the ON case at  $U = 20 \text{ m s}^{-1}$  did not converge.

All converged responses remain damped over the investigated speeds and initial conditions. No stable LCO branch or onset is obtained, so the present UVLM does not reproduce the experimental Fig. 8 bifurcation diagram.

García Pérez *et al.* associate the low-amplitude experimental branch with coupled-mode flutter and describe the subsequent larger-amplitude regime as stall flutter; the largest OFF responses are reported as most likely associated with dynamic stall flutter (García Pérez *et al.*, 2025). The present potential-flow UVLM contains no separated-flow or dynamic-stall model and is therefore not expected to reproduce that high-amplitude branch quantitatively. This limitation does not by itself explain the absence of the lower-amplitude flutter onset in the converged numerical calculations.

#### 5.2.4 Scope and limitations

The reference aerodynamic benchmarks are reproduced within the reported errors. The inferred-balanced reconstruction reproduces the OFF zero-flow frequencies, while the ON flap-dominated mode remains discrepant; the UVLM also does not reproduce the measured Fig. 7 branch evolution or yield a stable LCO in the converged Fig. 8 simulations.

The model assumes incompressible, inviscid potential flow with attached-flow loads and a prescribed non-roll-up wake, with no separated-flow or dynamic- stall model. The large-amplitude stall/dynamic-stall regime is therefore outside its expected scope, but this does not by itself explain the missing low-amplitude onset. Incomplete mounted-assembly and moving-flap inertias, damping, and raw Fig. 7–8 data, together with the shorter numerical histories, prevent attribution of the remaining disagreement to a unique source.

## CHAPTER 6 CONCLUSION

### 6.1 Synthesis of Work

The Euler phase established a complete two-dimensional solver and verification workflow. Its benchmark agreement is satisfactory, while the results show that residual convergence, coefficient stability, and asymptotic grid convergence must be distinguished.

The main scientific outcome is the verified low-fidelity aeroelastic framework for the partial-span flap–NES configuration. Its components reproduce the analytical/reference aerodynamic tests, but this component-level verification does not constitute application-level validation: converged UVLM simulations do not reproduce the frequency evolution or stable LCO branch reported by García Pérez et al.

The remaining discrepancy cannot be assigned to a unique cause. In particular, the absence of separated-flow and dynamic-stall modelling places the large-amplitude stall-flutter regime outside the expected scope of the present UVLM, but it does not by itself explain the missing low-amplitude flutter onset.

### 6.2 Limitations of the Proposed Solution

The comparison is limited by inferred structural mass and inertia properties, unreported damping, digitized experimental figures rather than raw data, shorter numerical histories, potential/inviscid/incompressible attached-flow assumptions, a prescribed non-roll-up wake, the absence of a separated-flow model, and some FSI cases that do not pass the convergence gate. The Euler results additionally retain finite oscillatory residuals on the finest subsonic grids and no clean single-term asymptotic regime for the finest transonic lifting sequence.

### 6.3 Future Work

Future work should first identify the mounted-assembly and flap mass properties, damping, and raw experimental histories. An independent UVLM, laboratory solver, or harmonic-balance model would help separate structural reconstruction uncertainty from aerodynamic-operator effects. Efficient nonlinear corrections such as NLUVLM or polar-based models, together with a controlled free-wake sensitivity study, could address larger-amplitude separated-flow behaviour; neither should be assumed to recover the low-amplitude onset. Selective higher-fidelity cases should remain diagnostics, not replace the reduced model; Euler development should prioritize residual robustness and profiling-guided performance.

## REFERENCES

- AYALA, S., “Harmonic balance vortex lattice method with numerical continuation for nonlinear aeroelastic analysis,” Master’s thesis, Polytechnique Montréal, 2025. [Online]. Available: <https://publications.polymtl.ca/71958/>
- BLAZEK, J., *Computational Fluid Dynamics: Principles and Applications:(Book with accompanying CD)*. Elsevier, 2005.
- GARCÍA PÉREZ, J., SANCHES, L., GHADAMI, A., EPUREANU, B. I., and MICHON, G., “Experimental parametric study of a flap-NES passive absorber for post-flutter control,” *Journal of Fluids and Structures*, vol. 138, p. 104405, 2025.
- GARRIGUES, E., “A review of industrial aeroelasticity practices at dassault aviation for military aircraft and business jets,” *AerospaceLab*, no. 14, September 2018, online journal. [Online]. Available: [https://aerospacelab.onera.fr/sites/w3.onera.fr.aerospacelab/files/AL14-09\\_0\\_0.pdf](https://aerospacelab.onera.fr/sites/w3.onera.fr.aerospacelab/files/AL14-09_0_0.pdf)
- JONES, R. T., “Operational treatment of the nonuniform-lift theory in airplane dynamics,” National Advisory Committee for Aeronautics, NACA Technical Report, 1938. [Online]. Available: <https://ntrs.nasa.gov/search.jsp?R=19930081472>
- NEWMARK, N. M., “A method of computation for structural dynamics,” *Journal of the Engineering Mechanics Division*, vol. 85, no. EM3, pp. 67–94, 1959.
- PARENTEAU, M., “A non-linear frequency domain potential flow model for compressible, transonic and viscous aeroelastic analyses,” Ph.D. dissertation, Polytechnique Montréal, 2021. [Online]. Available: <https://publications.polymtl.ca/6629/>
- SWANSON, R. R. and TURKEL, E., “Multistage schemes with multigrid for euler and navier-stokes equations,” *NASA Technical Paper*, no. 3631, 1997.
- THEODORSEN, T., “General theory of aerodynamic instability and the mechanism of flutter,” National Advisory Committee for Aeronautics, NACA Technical Report 496, 1949. [Online]. Available: <https://ntrs.nasa.gov/citations/19930090935>
- VASSBERG, J. C. and JAMESON, A., “In pursuit of grid convergence for two-dimensional euler solutions,” *Journal of Aircraft*, vol. 47, no. 4, pp. 1150–1166, 2010, originally presented as AIAA Paper 2009-4114.

## APPENDIX A EULER SOLVER TECHNICAL AND NUMERICAL DETAILS

### A.1 Finite-Volume Geometry and Boundary Details

The structured O-mesh is composed of quadrilateral cells whose conservative state is stored at the cell centre. For a cell with ordered vertices, the signed area and physical area are

$$A_s = \frac{1}{2} \sum_{k=1}^4 (x_k y_{k+1} - y_k x_{k+1}), \quad \Omega_{ij} = |A_s|, \quad (x_5, y_5) = (x_1, y_1). \quad (\text{A.1})$$

The area centroid is

$$(x_c, y_c) = \frac{1}{6A_s} \sum_{k=1}^4 (x_k + x_{k+1}, y_k + y_{k+1})(x_k y_{k+1} - x_{k+1} y_k). \quad (\text{A.2})$$

For a face joining vertices  $a$  and  $b$ , the candidate vector and midpoint are

$$\mathbf{s}_{ab}^{(0)} = \begin{bmatrix} y_b - y_a \\ -(x_b - x_a) \end{bmatrix}, \quad \mathbf{x}_f = \frac{1}{2}(\mathbf{x}_a + \mathbf{x}_b). \quad (\text{A.3})$$

The sign of  $\mathbf{s}_{ab}^{(0)}$  is selected from  $(\mathbf{x}_f - \mathbf{x}_c) \cdot \mathbf{s}_{ab}^{(0)}$  so that  $\mathbf{s}_f$  is outward. Hence  $\ell_f = \|\mathbf{s}_f\|_2$  and  $\mathbf{n}_f = \mathbf{s}_f / \ell_f$ . The four faces are ordered as radial inner, circumferential plus, radial outer, and circumferential minus. The circumferential index is periodic and the radial index is bounded.

### A.2 JST / Swanson–Tukel Boundary and Stencil Details

For direction  $d \in \{i, j\}$ , with  $d^\perp$  denoting the transverse direction, the cell spectral scales are

$$\lambda_{C,d} = (|\mathbf{u}_C \cdot \mathbf{n}_{C,d}| + a_C) \ell_{C,d}, \quad \Lambda_{C,d} = \lambda_{C,d} + \sqrt{\lambda_{C,d} \lambda_{C,d^\perp}}, \quad \Lambda_{r+1/2,d} = \frac{1}{2}(\Lambda_{r,d} + \Lambda_{r+1,d}). \quad (\text{A.4})$$

For the normal convective Jacobian

$$\mathbf{A}_n = \frac{\partial \mathbf{F}_n}{\partial \mathbf{U}}, \quad \text{eig}(\mathbf{A}_n) = \{u_n - a, u_n, u_n, u_n + a\}, \quad \rho(\mathbf{A}_n) = |u_n| + a. \quad (\text{A.5})$$

Thus, the scalar factor used in the present JST dissipation is the Euler normal spectral radius, augmented by the face measure and the documented transverse scaling. It multiplies

the state-difference vector as a scalar. A matrix-dissipation formulation would instead use

$$|\mathbf{A}_n| = \mathbf{R} |\mathbf{\Lambda}_A| \mathbf{R}^{-1}, \quad (\text{A.6})$$

so that the characteristic modes receive dissipation associated with their individual wave speeds. The present solver does not use this matrix form; the distinction is the one made between scalar JST spectral-radius scaling and matrix dissipation in the Swanson–Tukel discussion (Swanson and Tukel, 1997). The JST stencil wraps in the circumferential direction. In the radial direction the index is clipped and ghost states close the state and pressure stencils. The fourth-difference contribution is suppressed on the two internal radial faces adjacent to each physical boundary. The physical wall and far-field JST fluxes are zero. On coarse multigrid levels, the first-order branch uses  $k_2 = 0.5$ ,  $k_4 = 0$  and

$$\mathbf{Q}_f^{(1)} = \frac{1}{2} \lambda_f^{\text{avg}} \ell_f (\mathbf{U}_R - \mathbf{U}_L), \quad \lambda_f^{\text{avg}} = \frac{1}{2} [(|u_{n,L}| + a_L) + (|u_{n,R}| + a_R)]. \quad (\text{A.7})$$

For an interior face, the centred flux and total JST flux are related by

$$\Phi_f^c = \frac{\ell_f}{2} [\mathbf{F}_n(\mathbf{U}_L) + \mathbf{F}_n(\mathbf{U}_R)], \quad \Phi_f = \Phi_f^c - \mathbf{Q}_f. \quad (\text{A.8})$$

At a wall, the pressure used by the centred wall flux is extrapolated from the first two cell centres. With  $\mathbf{n}_w$  the outward wall normal,  $d_k = -(\mathbf{x}_{C_k} - \mathbf{x}_w) \cdot \mathbf{n}_w$ , and  $p_1, p_2$  the corresponding pressures,

$$p_w = \begin{cases} p_1 + d_1 \frac{p_1 - p_2}{d_2 - d_1}, & d_1 > 0, \ d_2 - d_1 > 10^{-12} \max(1, |d_1|, |d_2|), \\ p_1, & \text{otherwise,} \end{cases} \quad (\text{A.9})$$

$$\Phi_w = \ell_w \begin{bmatrix} 0 \\ p_w n_{x,w} \\ p_w n_{y,w} \\ 0 \end{bmatrix}, \quad \mathbf{Q}_w = \mathbf{0}.$$

The two wall ghost layers use the reflected primitive state defined in Eq. (2.14).

At the outer boundary,  $\mathbf{n}$  is the outward normal of the fluid domain. Define

$$u_{n,i} = \mathbf{u}_i \cdot \mathbf{n}, \quad u_{n,\infty} = \mathbf{u}_\infty \cdot \mathbf{n}, \quad M_i = \frac{\|\mathbf{u}_i\|_2}{a_i}. \quad (\text{A.10})$$

Supersonic inflow ( $M_i \geq 1$ ,  $u_{n,i} < 0$ ) uses the corrected far-field state, whereas supersonic

outflow uses the interior state. For subsonic inflow, the characteristic state is

$$\begin{aligned} p_b &= \frac{1}{2} [p_\infty + p_i - \rho_i a_i (u_{n,\infty} - u_{n,i})], & \rho_b &= \rho_\infty + \frac{p_b - p_\infty}{a_\infty^2}, \\ u_{n,b} &= u_{n,\infty} + \frac{p_b - p_\infty}{\rho_\infty a_\infty}, & u_{t,b} &= u_{t,\infty}. \end{aligned} \quad (\text{A.11})$$

For subsonic outflow,

$$p_b = p_\infty, \quad \rho_b = \rho_i + \frac{p_\infty - p_i}{a_i^2}, \quad u_{n,b} = u_{n,i} + \frac{p_i - p_\infty}{\rho_i a_i}, \quad u_{t,b} = u_{t,i}. \quad (\text{A.12})$$

The normal and tangential components are transformed back to  $(u_b, v_b)$ . The two far-field ghost layers are extrapolated from  $W_b$  as  $W_{h1} = 2W_b - W_i$  and  $W_{h2} = 2W_{h1} - W_i$ ; a non-physical extrapolation is replaced successively by  $W_b$  and  $W_{h1}$ .

For the point-vortex correction used in lifting calculations, the circulation is updated after the residual evaluation and before the Runge–Kutta sweep. In a multigrid run, this update is made once per cycle and synchronized across levels.

### A.3 FAS / FMG Transfer Operators and Cycle Details

The CFL number can be ramped according to

$$\text{CFL}_n = \begin{cases} \text{CFL}_0 + \min(1, n/N_{\text{ramp}})(\text{CFL}_{\text{max}} - \text{CFL}_0), & N_{\text{ramp}} > 0, \\ \text{CFL}_{\text{max}}, & N_{\text{ramp}} = 0. \end{cases} \quad (\text{A.13})$$

Within a Runge–Kutta sweep, the JST contribution is frozen except for the stage-one recomputation following a relaxed vortex-circulation update. On a forced FAS level, the forcing is subtracted from the residual before smoothing.

When IRS is enabled, set  $q_{ij}^{(0)} = \Delta t_{ij}(R_{ij} - f_{ij})$  and  $\epsilon_i = \epsilon_j = \epsilon_{\text{IRS}}$ . In the periodic direction, each component satisfies

$$-\epsilon_i q_{i-1,j} + (1 + 2\epsilon_i) q_{i,j} - \epsilon_i q_{i+1,j} = q_{i,j}^{(0)}, \quad i \bmod N_i. \quad (\text{A.14})$$

In the bounded direction,

$$-\epsilon_j q_{i,j-1} + d_j q_{i,j} - \epsilon_j q_{i,j+1} = q_{i,j}^{(0)}, \quad d_j = \begin{cases} 1 + \epsilon_j, & j = j_s \text{ or } j = j_e - 1, \\ 1 + 2\epsilon_j, & j_s < j < j_e - 1. \end{cases} \quad (\text{A.15})$$

The smoothed residual is  $\tilde{R}_{ij} = q_{ij}/\Delta t_{ij}$ . A cyclic solve in  $i$  is followed by a Thomas solve in

$j$ , with one factorization per Runge–Kutta stage.

The hierarchy halves both mesh dimensions and retains coarse nodes at  $\mathbf{x}_{I,J}^{\ell+1} = \mathbf{x}_{2I,2J}^\ell$ . Area-weighted state restriction is

$$\mathbf{U}_{I,J}^H = R_A \mathbf{U}^h = \frac{\sum_{\delta_i, \delta_j=0}^1 \Omega_{2I+\delta_i, 2J+\delta_j}^h \mathbf{U}_{2I+\delta_i, 2J+\delta_j}^h}{\sum_{\delta_i, \delta_j=0}^1 \Omega_{2I+\delta_i, 2J+\delta_j}^h}. \quad (\text{A.16})$$

For  $\mathbf{r}^h = L^h(\mathbf{U}^h) - \mathbf{f}^h$ , residual restriction and FAS forcing are

$$\mathbf{r}_{I,J}^{H,\text{res}} = \frac{1}{\Omega_{I,J}^H} \sum_{\delta_i, \delta_j=0}^1 \Omega_{2I+\delta_i, 2J+\delta_j}^h \mathbf{r}_{2I+\delta_i, 2J+\delta_j}^h, \quad \mathbf{f}^H = L^H(R_A \mathbf{U}^h) - R_A \mathbf{r}^h. \quad (\text{A.17})$$

The coarse equation is  $L^H(\mathbf{U}^H) = \mathbf{f}^H$ . The coarse-level dissipation uses  $k_2 = 0.5$ ,  $k_4 = 0$ ; final cycles use the standard JST branch on the fine root and first-order dissipation on configured descendants.

For child  $(2I + d_i, 2J + d_j)$ , bilinear prolongation of a coarse correction is

$$P_{bil} \delta \mathbf{U}^H = \frac{9}{16} \delta \mathbf{U}_{I,J}^H + \frac{3}{16} \delta \mathbf{U}_{I_n,J}^H + \frac{3}{16} \delta \mathbf{U}_{I,J_n}^H + \frac{1}{16} \delta \mathbf{U}_{I_n,J_n}^H, \quad (\text{A.18})$$

where  $I_n = I - 1$  for  $d_i = 0$  and  $I_n = I + 1$  for  $d_i = 1$ , with the analogous definition for  $J_n$ . The periodic index is wrapped and the bounded index is clipped. The fine state is corrected by

$$\mathbf{U}^h \leftarrow \mathbf{U}^h + \omega_\ell P_{bil} \delta \mathbf{U}^H. \quad (\text{A.19})$$

The cycle index is  $\gamma = 1$  for a V-cycle and  $\gamma = 2$  for a W-cycle. FMG initialization proceeds from coarse to fine, with first-order dissipation on the coarsest level before the intermediate FAS cycles.

#### A.4 Component Verification Tests

The automated checks reported for the Euler solver are:

1. constant-state geometric residual below  $10^{-8}$ ;
2. full residual for  $\mathbf{W} = (1, 0, 0, 1)$  below  $10^{-8}$ ;
3. reflected wall-state consistency;

4. preservation of  $h_{0,\Gamma} = h_{0,\infty}$  and  $p_\Gamma/\rho_\Gamma^\gamma = p_\infty/\rho_\infty^\gamma$  for the vortex state;
5.  $\Delta t = \text{CFL}/(4a)$  on a unit resting Cartesian cell;
6. activation of the second-difference term on the first radial face; and
7. agreement between frozen-JST and recomputed-JST residuals.

The automated Euler test suite passes. A global conservation diagnostic, a zero-angle lift assertion, dedicated IRS/FAS/prolongation tests, and automated order estimation are not claimed as completed tests.

For a mono-grid run, the implementation-level stopping expression is

$$\log_{10} \left( \max \left( \frac{\|\mathbf{R}^n\|_\infty}{R^0}, \text{DBL\_MIN} \right) \right) \leq -\text{convergence\_orders}.$$

For multigrid, the active criterion is  $\|\mathbf{R}^n\|_\infty \leq 10^{-6}$ ; stopping is not based on changes in  $C_L$ ,  $C_D$ , or  $C_M$ .

## A.5 Complete Grid-Convergence Data

The following tables report the complete coefficient sequences used in Chapter 3. Values are retained at the precision needed to reproduce the reported grid differences; quasi-zero lift and moment values reflect the symmetry level of the nonlifting calculations.

Table A.1 Complete grid-convergence coefficients for the subsonic nonlifting case ( $M_\infty = 0.50$ ,  $\alpha = 0^\circ$ ).

| $N$  | $C_L$                                | $C_D$                   | $C_M$                                |
|------|--------------------------------------|-------------------------|--------------------------------------|
| 32   | $-1.903239470785983 \times 10^{-15}$ | 0.004473360921115549    | $-7.781473877953107 \times 10^{-16}$ |
| 64   | $-3.318278191904207 \times 10^{-15}$ | 0.000579960977517635    | $1.231653667943533 \times 10^{-15}$  |
| 128  | $-3.899286647536725 \times 10^{-14}$ | $-0.000051136124109547$ | $8.741147686627419 \times 10^{-15}$  |
| 256  | $3.835346598844515 \times 10^{-13}$  | $-0.000050865578732283$ | $-9.180280543688976 \times 10^{-14}$ |
| 512  | $-2.192995986318288 \times 10^{-12}$ | $-0.000018457508440850$ | $5.109969522174018 \times 10^{-13}$  |
| 1024 | $1.089614477785310 \times 10^{-11}$  | $-0.000005411960078842$ | $-2.536841092708162 \times 10^{-12}$ |
| 2048 | $8.846850721647047 \times 10^{-11}$  | $-0.000001461985498501$ | $8.129849925724150 \times 10^{-12}$  |

Table A.2 Complete grid-convergence coefficients for the subsonic lifting case ( $M_\infty = 0.50$ ,  $\alpha = 1.25^\circ$ ).

| $N$  | $C_L$             | $C_D$              | $C_M$              |
|------|-------------------|--------------------|--------------------|
| 32   | 0.177619899816708 | 0.004942918269659  | -0.003563492753445 |
| 64   | 0.178545538036491 | 0.000776873996082  | -0.002427168894166 |
| 128  | 0.179561655388539 | -0.000017745449201 | -0.002209852401777 |
| 256  | 0.179950406713433 | -0.000047185575124 | -0.002204029825194 |
| 512  | 0.180111677678676 | -0.000017984636889 | -0.002223740982642 |
| 1024 | 0.180206213343130 | -0.000005027073096 | -0.002242501288005 |
| 2048 | 0.180262647159837 | -0.000000982207620 | -0.002255076749653 |

Table A.3 Complete grid-convergence coefficients for the transonic nonlifting case ( $M_\infty = 0.80$ ,  $\alpha = 0^\circ$ ).

| $N$  | $C_L$                                | $C_D$             | $C_M$                                |
|------|--------------------------------------|-------------------|--------------------------------------|
| 32   | $2.981555974335137 \times 10^{-16}$  | 0.012326767376398 | $7.318364664277155 \times 10^{-16}$  |
| 64   | $-1.116921845190871 \times 10^{-14}$ | 0.008534202984368 | $2.216806227671255 \times 10^{-15}$  |
| 128  | $-9.064196372333668 \times 10^{-14}$ | 0.008209705464094 | $2.174575585015290 \times 10^{-14}$  |
| 256  | $9.656987540281320 \times 10^{-13}$  | 0.008274532092860 | $-2.320247706260551 \times 10^{-13}$ |
| 512  | $-4.500522053317026 \times 10^{-12}$ | 0.008319492966186 | $1.083803401474408 \times 10^{-12}$  |
| 1024 | $2.073701295773231 \times 10^{-11}$  | 0.008335511055792 | $-4.999066141624738 \times 10^{-12}$ |
| 2048 | $1.706838406172179 \times 10^{-10}$  | 0.008340262540741 | $-4.115215128548840 \times 10^{-11}$ |

Table A.4 Complete grid-convergence coefficients for the transonic lifting case ( $M_\infty = 0.80$ ,  $\alpha = 1.25^\circ$ ).

| $N$  | $C_L$             | $C_D$             | $C_M$              |
|------|-------------------|-------------------|--------------------|
| 32   | 0.357548672208026 | 0.027209980365859 | -0.038267919640774 |
| 64   | 0.367916383070424 | 0.023907498632738 | -0.042088299636375 |
| 128  | 0.366614800826015 | 0.023187042687106 | -0.041782254273207 |
| 256  | 0.362624349277871 | 0.022983668444203 | -0.040672568001944 |
| 512  | 0.359421110047434 | 0.022844912478189 | -0.039752899176762 |
| 1024 | 0.358283884050049 | 0.022795325518665 | -0.039426218927689 |
| 2048 | 0.356574824105996 | 0.022706737060007 | -0.038928963279638 |

## A.6 Iterative Convergence and Solver Performance

The multigrid-versus-single-grid residual histories are retained in the main text because they directly support the iterative-convergence discussion. The following timing comparison is kept in the appendix because it measures implementation-level scaling rather than a change in the numerical solution.

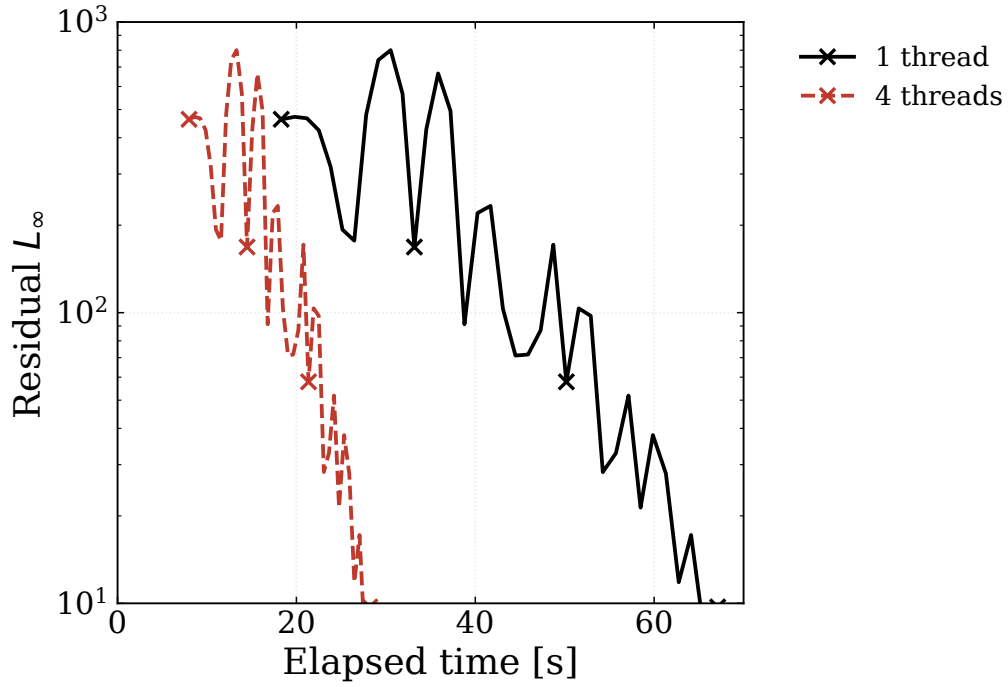


Figure A.1 Single-thread and four-thread timing comparison for the  $512^2$  case.

## APPENDIX B AEROELASTIC TECHNICAL DETAILS

### B.1 Structural scaling and parameter provenance

The structural equations in the main text use the dimensional coordinate vector  $\mathbf{q} = \mathbf{D}\mathbf{x}$ , with  $\mathbf{D} = \text{diag}(b, 1, 1)$ . Since  $t = \tau/\omega_\alpha$ , multiplication of the dimensional equation by  $\mathbf{D}^T/(mb^2\omega_\alpha^2)$  gives

$$\mathbf{M}_s = \frac{\mathbf{D}^T \mathbf{M}_{\text{dim}} \mathbf{D}}{mb^2}, \quad \mathbf{C}_s = \frac{\mathbf{D}^T \mathbf{C}_{\text{dim}} \mathbf{D}}{mb^2\omega_\alpha}, \quad \mathbf{K}_s = \frac{\mathbf{D}^T \mathbf{K}_{\text{dim}} \mathbf{D}}{mb^2\omega_\alpha^2}. \quad (\text{B.1})$$

The dimensionless mass parameters used in the structural initialization are

$$x_\alpha = \frac{S_\alpha}{mb}, \quad x_\beta = \frac{S_\beta}{mb}, \quad r_\alpha^2 = \frac{I_\alpha}{mb^2}, \quad r_\beta^2 = \frac{I_\beta}{mb^2}, \quad \sigma = \frac{\omega_h}{\omega_\alpha}. \quad (\text{B.2})$$

For a released flap, the matrices assembled by the present dimensional-law path are therefore

$$\mathbf{M}_s = \begin{bmatrix} m_t/m & x_\alpha & x_\beta \\ x_\alpha & r_\alpha^2 & (c_h - a)x_\beta + r_\beta^2 \\ x_\beta & (c_h - a)x_\beta + r_\beta^2 & r_\beta^2 \end{bmatrix}, \quad (\text{B.3})$$

$$\mathbf{C}_s = 2 \text{diag} \left( (m_t/m)\sigma\zeta_h, r_\alpha^2\zeta_\alpha, (\omega_\beta/\omega_\alpha)r_\beta^2\zeta_\beta \right), \quad (\text{B.4})$$

$$\mathbf{K}_s = \text{diag} \left( \frac{k_h}{m\omega_\alpha^2}, \frac{k_{\alpha 1}}{mb^2\omega_\alpha^2}, \frac{k_{\beta 1}}{mb^2\omega_\alpha^2} \right). \quad (\text{B.5})$$

The nonlinear contribution in the same scaling is

$$\mathbf{f}_{\text{nl}}^*(\mathbf{x}) = \begin{bmatrix} 0 \\ k_{\alpha 3}\alpha^3/(mb^2\omega_\alpha^2) \\ k_{\beta 3}\beta^3/(mb^2\omega_\alpha^2) \end{bmatrix}. \quad (\text{B.6})$$

The fixed three-by-three storage layout is retained for both configurations. For a locked flap, the third structural mass entry is a unit placeholder and the third state, force, and nonlinear contribution are set to zero; the physical equation is the active  $(h, \alpha)$  restriction.

The following table records the origin of the principal quantities. “Published” means directly stated in the Michon article. “Configuration” means an input used by the present solver but not claimed here as a measured article value. “Estimate” means that the quantity is derived from other inputs.

Table B.1 Provenance of structural quantities used by the Michon path.

| Quantity                             | Category           | Present interpretation                                                                |
|--------------------------------------|--------------------|---------------------------------------------------------------------------------------|
| $m_w = 2.4$ kg                       | Published          | Reported wing assembly mass.                                                          |
| $m_{add} = 0.04$ kg                  | Published          | Added mass for the Fig. 7 hardware, placed 0.08 m aft of the main trailing edge.      |
| $k_h, k_{\alpha 1}, k_{\alpha 3}$    | Published          | Plunge stiffness and pitch restoring-law coefficients.                                |
| $k_{\beta 1}, k_{\beta 3}$           | Published          | SPF1 flap restoring-law coefficients.                                                 |
| $\zeta_h, \zeta_\alpha, \zeta_\beta$ | Configuration      | Temporary input values marked as not identified from the Michon article.              |
| $I_\alpha, S_\alpha$                 | Estimate or preset | Constructed from the selected mass-property preset; not silently treated as measured. |
| $I_\beta, S_\beta$                   | Estimate           | Built from the available flap geometry and the added point mass.                      |

For the non-inferred baseline mass-property construction, the pitch inertia of the base 2.4 kg assembly is estimated with a uniform chordwise distribution,

$$I_{\alpha, \text{base}} = m_w \left( \frac{c^2}{12} + \left( \frac{c}{2} - x_{EA} \right)^2 \right). \quad (\text{B.7})$$

The added point mass contributes using the radius  $r_{add} = c + 0.08 - x_{EA}$ . Hence

$$I_\alpha = I_{\alpha, \text{base}} + m_{add} r_{add}^2, \quad S_\alpha = m_w \left( \frac{c}{2} - x_{EA} \right) + m_{add} r_{add}. \quad (\text{B.8})$$

For the flap estimate, the model uses the areal density of the reported wing assembly to assign a base flap mass  $m_f$ , a centroid offset  $d_f = 0.0175$  m from the hinge, and the added-mass radius  $r_f = 0.08 + 0.045 = 0.125$  m. The corresponding quantities are

$$S_\beta = m_f d_f + m_{add} r_f, \quad I_\beta = m_f \left( \frac{c_f^2}{12} + d_f^2 \right) + m_{add} r_f^2. \quad (\text{B.9})$$

The inferred-balanced preset replaces the base values in Eq. (B.8) by the explicit hypothesis  $S_{\alpha, \text{base}} = 0$  and  $I_{\alpha, \text{base}} = 0.014$  kg m<sup>2</sup>. This is an inverse modelling hypothesis and must not be described as a published experimental measurement.

## B.2 Jones–Theodorsen matrix realization

The reduced Jones model uses  $a = x_{EA}/b - 1$  and  $c_h = x_h/b - 1$ . Define

$$s_c = \sqrt{1 - c_h^2}, \quad \chi = \arccos(c_h), \quad (\text{B.10})$$

and the geometric functions

$$\begin{aligned} T_1 &= -\frac{1}{3}s_c(2 + c_h^2) + c_h\chi, \\ T_2 &= c_h(1 - c_h^2) - s_c(1 + c_h^2)\chi + c_h\chi^2, \\ T_3 &= -\left(\frac{1}{8} + c_h^2\right)\chi^2 + \frac{1}{4}c_h s_c\chi(7 + 2c_h^2) - \frac{1}{8}(1 - c_h^2)(5c_h^2 + 4), \\ T_4 &= -\chi + c_h s_c, \quad T_5 = -(1 - c_h^2) - \chi^2 + 2c_h s_c\chi, \\ T_6 &= T_2, \quad T_7 = -\left(\frac{1}{8} + c_h^2\right)\chi + \frac{1}{8}c_h s_c(7 + 2c_h^2), \\ T_8 &= -\frac{1}{3}s_c(2c_h^2 + 1) + c_h\chi, \\ T_9 &= \frac{1}{2}\left(\frac{1}{3}s_c^3 + aT_4\right), \\ T_{10} &= s_c + \chi, \\ T_{11} &= \chi(1 - 2c_h) + s_c(2 - c_h), \\ T_{12} &= s_c(2 + c_h) - \chi(2c_h + 1), \\ T_{13} &= \frac{1}{2}[-T_7 - (c_h - a)T_1]. \end{aligned} \quad (\text{B.11})$$

The aerodynamic inertia matrix is

$$\mathbf{M}_a = \begin{bmatrix} -1 & a & T_1/\pi \\ a & -(1/8 + a^2) & -2T_{13}/\pi \\ T_1/\pi & -2T_{13}/\pi & T_3/\pi^2 \end{bmatrix}. \quad (\text{B.12})$$

Before multiplication by  $V$ , the damping matrix is

$$\mathbf{D}_{a0} = \begin{bmatrix} -2 & -2(1-a) & (T_4 - T_{11})/\pi \\ 1+2a & a(1-2a) & [T_8 - T_1 + (c_h - a)T_4 + aT_{11}]/\pi \\ -T_{12}/\pi & [2T_9 + T_1 + (T_{12} - T_4)(a - 1/2)]/\pi & (T_{11}(T_4 - T_{12}))/(\pi^2) \end{bmatrix}, \quad (\text{B.13})$$

$$\mathbf{D}_a = V\mathbf{D}_{a0}.$$

Similarly,

$$\mathbf{K}_a = V^2 \begin{bmatrix} 0 & -2 & -2T_{10}/\pi \\ 0 & 1 + 2a & (2aT_{10} - T_4)/\pi \\ 0 & -T_{12}/\pi & -[T_5 - T_{10}(T_4 - T_{12})]/\pi^2 \end{bmatrix}. \quad (\text{B.14})$$

With  $d_1 = 0.165$ ,  $d_2 = 0.335$ ,  $\ell_1 = 0.0455$ , and  $\ell_2 = 0.300$ , the lag and kinematic matrices used in the reduced model are written with a signed and scaled lag-state convention. Consequently, the positive  $d_j$  entries below are consistent with the negative coefficients in the Wagner approximation of Eq. (4.11).

$$\mathbf{L}_\delta = 2V \begin{bmatrix} d_1 & d_2 \\ -(1/2 + a)d_1 & -(1/2 + a)d_2 \\ T_{12}d_1/(2\pi) & T_{12}d_2/(2\pi) \end{bmatrix}, \quad (\text{B.15})$$

$$\mathbf{Q}_{\lambda a} = \begin{bmatrix} 1 & 1/2 - a & T_{11}/(2\pi) \\ 1 & 1/2 - a & T_{11}/(2\pi) \end{bmatrix}, \quad (\text{B.16})$$

$$\mathbf{Q}_{\lambda v} = V \begin{bmatrix} 0 & 1 & T_{10}/\pi \\ 0 & 1 & T_{10}/\pi \end{bmatrix}, \quad \mathbf{L}_\lambda = V \text{diag}(-\ell_1, -\ell_2). \quad (\text{B.17})$$

The matrices above are the explicit realization of Eq. (4.13). The first matrix multiplies the generalized accelerations and the second multiplies the generalized velocities in the lag-state equation. They apply to the reduced sectional model; they are not used to infer the circulation or wake state of the multi-surface UVLM path.

### B.3 UVLM discrete assembly and wake bookkeeping

For a finite vortex segment with endpoints  $\mathbf{x}_1$  and  $\mathbf{x}_2$ , let

$$\mathbf{r}_0 = \mathbf{x}_2 - \mathbf{x}_1, \quad \mathbf{r}_1 = \mathbf{x} - \mathbf{x}_1, \quad \mathbf{r}_2 = \mathbf{x} - \mathbf{x}_2. \quad (\text{B.18})$$

The numerical kernel uses the regularized finite-segment form

$$\mathbf{v}_{seg}(\mathbf{x}) = \frac{\mathbf{r}_1 \times \mathbf{r}_2}{4\pi r_1 r_2} \frac{\mathbf{r}_0 \cdot \mathbf{r}_1 r_2 - \mathbf{r}_0 \cdot \mathbf{r}_2 r_1}{\sqrt{\|\mathbf{r}_1 \times \mathbf{r}_2\|^4 + \sigma_v^4 \|\mathbf{r}_0\|^4}}, \quad (\text{B.19})$$

where  $r_j = \|\mathbf{r}_j\|$  and  $\sigma_v$  is the kernel regularization parameter. A quadrilateral ring contribution is the sum of the four segment contributions, multiplied by its circulation. Optional spanwise mirror symmetry is applied by adding the mirrored segment velocity when enabled by the numerical formulation.

The right-hand side is assembled from the panel velocity and wake influence,

$$b_i = -\mathbf{V}_{\text{body},i} \cdot \mathbf{n}_i - \mathbf{V}_{\text{wake},i} \cdot \mathbf{n}_i, \quad (\text{B.20})$$

and the bound influence matrix contains the normal projection of every bound ring at every collocation point. Wake influence is evaluated directly from wake vertices and circulation instead of forming a separate dense wake matrix. The global bound system is assembled over the ordered list of configured surfaces.

At a time-level update, the wake vertex station is assigned from the trailing edge of the current geometry,

$$\mathbf{X}_w^{j+1}(:, 0) = \mathbf{X}_{TE}^{j+1}. \quad (\text{B.21})$$

For a grouped partial-span wake, the main aft strips and flap trailing edge are concatenated into a single spanwise wake carrier. At a nonzero relative flap angle, the geometric interface vertex used by the wake topology is the midpoint of the adjacent trailing-edge vertices. This bookkeeping preserves the configured wake connectivity without introducing a free-wake convection step.

The chordwise circulation jump used by the force kernel follows Eq. (4.21). The circulation derivative is a backward time difference over the accepted step,

$$\dot{\Gamma}_{i,j}^{n+1} = \frac{\Gamma_{i,j}^{n+1} - \Gamma_{i,j}^n}{\Delta\tau}. \quad (\text{B.22})$$

The force evaluation then forms the working-density force in Eq. (4.22). The density and dimensional factors are introduced only by the generalized-load reduction,

$$\eta = \frac{\rho b^3}{m}, \quad \mathbf{Q}_a^* = \eta \begin{bmatrix} -\sum_i f_{i,z} \\ \sum_i (\mathbf{r}_{EA,i} \times \mathbf{f}_i)_y \\ \sum_{i \in \text{flap}} (\mathbf{r}_{h,i} \times \mathbf{f}_i)_y \end{bmatrix}. \quad (\text{B.23})$$

### B.3.1 Hierarchical rigid-body kinematics

The TinyVLM kinematic interface represents each node by a  $4 \times 4$  dual homogeneous transformation. Introduce the dual time variable

$$\hat{t} = t + \varepsilon, \quad \varepsilon^2 = 0, \quad f(\hat{t}) = f(t) + \varepsilon \dot{f}(t). \quad (\text{B.24})$$

The value part of a node transform has the homogeneous form

$$\mathbf{T}_i(t) = \begin{bmatrix} \mathbf{R}_i(t) & \boldsymbol{\tau}_i(t) \\ \mathbf{0}^T & 1 \end{bmatrix}, \quad (\text{B.25})$$

where  $\mathbf{R}_i$  is the rotational block and  $\boldsymbol{\tau}_i$  is the translation block. The kinematic implementation stores these matrices column-wise and multiplies them by homogeneous column vectors. Its pure translation constructor is therefore

$$\mathbf{T}_{\text{tr}}(\mathbf{d}) = \begin{bmatrix} \mathbf{I} & \mathbf{d} \\ \mathbf{0}^T & 1 \end{bmatrix}. \quad (\text{B.26})$$

For the unit axes used by the Michon configuration, the rotation constructor uses Rodrigues' expression

$$\mathbf{R}(\mathbf{a}, \theta) = \mathbf{I} + \sin \theta [\mathbf{a}]_{\times} + (1 - \cos \theta) [\mathbf{a}]_{\times}^2, \quad \boldsymbol{\tau} = (\mathbf{I} - \mathbf{R})\mathbf{p}, \quad (\text{B.27})$$

where  $\mathbf{p}$  is the point about which the rotation is performed. The supplied axis is the unit spanwise vector  $(0, 1, 0)$ , and the upper-right homogeneous block is generated by  $(\mathbf{I} - \mathbf{R})\mathbf{p}$ .

For a parent-child relation, the global transform is evaluated recursively as

$$\hat{\mathbf{T}}_{i,\text{global}} = \hat{\mathbf{T}}_{\text{parent,global}} \hat{\mathbf{T}}_{i,\text{local}}, \quad (\text{B.28})$$

with the local transform on the right. A material point therefore satisfies

$$\begin{bmatrix} \hat{\mathbf{x}}_g \\ 1 \end{bmatrix} = \hat{\mathbf{T}}_{\text{global}} \begin{bmatrix} \mathbf{x}_0 \\ 1 \end{bmatrix}, \quad \hat{\mathbf{x}}_g = \mathbf{x}_g + \varepsilon \dot{\mathbf{x}}_g. \quad (\text{B.29})$$

The linear velocity is the coefficient of  $\varepsilon$  in this product. For the angular velocity helper, the implementation extracts  $\mathbf{R}$  and  $\dot{\mathbf{R}}$  from the same dual transform, forms  $\boldsymbol{\Omega} = \dot{\mathbf{R}}\mathbf{R}^T$ , and returns the three skew components  $(\Omega_{yz}, \Omega_{zx}, \Omega_{xy})$ .

For the Michon specialization at evaluation time  $\tau_{n+1}$ , the local time argument is  $\ell = t - \tau_{n+1}$  and the formulation constructs

$$\mathbf{T}_m(t) = \mathbf{T}_{\text{tr}}(\mathbf{d}_m) \mathbf{R}_y (1 + a, \alpha + \dot{\alpha} \ell), \quad \mathbf{d}_m = \begin{bmatrix} 0 & 0 & -h - \dot{h} \ell \end{bmatrix}^T, \quad (\text{B.30})$$

$$\mathbf{T}_f(t) = \mathbf{R}_y (1 + c_h, \beta + \dot{\beta} \ell), \quad \mathbf{T}_{f,\text{global}} = \mathbf{T}_m \mathbf{T}_f. \quad (\text{B.31})$$

Here  $1 + a$  and  $1 + c_h$  are the nondimensional elastic-axis and flap-hinge locations. The main node is a direct child of the root and the flap node is a child of the main node; positions and

velocities are obtained from the same parent-child composition.

#### B.4 Newmark- $\beta$ integration

##### B.4.1 Kinematic assumptions

For a structural state at  $n$  and  $n + 1$ , the Newmark assumptions are

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta\tau \mathbf{x}'_n + \Delta\tau^2 \left[ \left( \frac{1}{2} - \beta_N \right) \mathbf{x}''_n + \beta_N \mathbf{x}''_{n+1} \right], \quad (\text{B.32})$$

$$\mathbf{x}'_{n+1} = \mathbf{x}'_n + \Delta\tau \left[ (1 - \gamma_N) \mathbf{x}''_n + \gamma_N \mathbf{x}''_{n+1} \right]. \quad (\text{B.33})$$

Solving the first relation for the new acceleration gives

$$\mathbf{x}''_{n+1} = a_0(\mathbf{x}_{n+1} - \mathbf{x}_n) - a_2 \mathbf{x}'_n - a_3 \mathbf{x}''_n, \quad (\text{B.34})$$

and substitution into the second relation gives

$$\mathbf{x}'_{n+1} = a_1(\mathbf{x}_{n+1} - \mathbf{x}_n) - a_4 \mathbf{x}'_n - a_5 \mathbf{x}''_n, \quad (\text{B.35})$$

with

$$a_0 = \frac{1}{\beta_N \Delta\tau^2}, \quad a_1 = \frac{\gamma_N}{\beta_N \Delta\tau}, \quad a_2 = \frac{1}{\beta_N \Delta\tau}, \quad (\text{B.36})$$

$$a_3 = \frac{1}{2\beta_N} - 1, \quad a_4 = \frac{\gamma_N}{\beta_N} - 1, \quad a_5 = \Delta\tau \left( \frac{\gamma_N}{2\beta_N} - 1 \right). \quad (\text{B.37})$$

##### B.4.2 Effective structural system

Substitution into

$$\mathbf{M}_s \mathbf{x}''_{n+1} + \mathbf{C}_s \mathbf{x}'_{n+1} + \mathbf{K}_s \mathbf{x}_{n+1} = \mathbf{Q}_{n+1} \quad (\text{B.38})$$

gives

$$\boxed{\mathbf{K}_{\text{eff}} = \mathbf{K}_s + a_0 \mathbf{M}_s + a_1 \mathbf{C}_s} \quad (\text{B.39})$$

and

$$\boxed{\mathbf{K}_{\text{eff}} \mathbf{x}_{n+1} = \mathbf{Q}_{n+1} + \mathbf{r}_n}, \quad (\text{B.40})$$

where the known-state contribution is

$$\mathbf{r}_n = \mathbf{M}_s (a_0 \mathbf{x}_n + a_2 \mathbf{x}'_n + a_3 \mathbf{x}''_n) + \mathbf{C}_s (a_1 \mathbf{x}_n + a_4 \mathbf{x}'_n + a_5 \mathbf{x}''_n). \quad (\text{B.41})$$

### B.4.3 Incremental form used by the solver

The time integrator solves for  $\Delta \mathbf{x}$ , with  $\mathbf{x}_{n+1} = \mathbf{x}_n + \Delta \mathbf{x}$ . Its coefficient names are

$$x_0 = a_0, \quad x_1 = a_1, \quad x_{d0} = a_2, \quad (\text{B.42})$$

$$x_{d1} = \frac{\gamma_N}{\beta_N} = a_4 + 1, \quad x_{dd0} = \frac{1}{2\beta_N} = a_3 + 1, \quad x_{dd1} = a_5. \quad (\text{B.43})$$

The last two shifts occur because the algorithm stores velocity and acceleration increments rather than absolute values. In particular,

$$\Delta \mathbf{x}' = x_1 \Delta \mathbf{x} - x_{d1} \mathbf{x}'_n - x_{dd1} \mathbf{x}''_n, \quad (\text{B.44})$$

$$\Delta \mathbf{x}'' = x_0 \Delta \mathbf{x} - x_{d0} \mathbf{x}'_n - x_{dd0} \mathbf{x}''_n. \quad (\text{B.45})$$

The linear system solved at each subiteration is

$$(\mathbf{K}_s + x_0 \mathbf{M}_s + x_1 \mathbf{C}_s) \Delta \mathbf{x} = \Delta \mathbf{F} + \mathbf{M}_s (x_{d0} \mathbf{x}'_n + x_{dd0} \mathbf{x}''_n) + \mathbf{C}_s (x_{d1} \mathbf{x}'_n + x_{dd1} \mathbf{x}''_n). \quad (\text{B.46})$$

The force increment passed to this system is, for the dimensional restoring-law path,

$$\Delta \mathbf{F}^{(k+1)} = \mathbf{Q}_a^{n+1,k+1} - \mathbf{Q}_a^n - [\mathbf{f}_{\text{nl}}^*(\mathbf{x}^{n+1,k+1}) - \mathbf{f}_{\text{nl}}^*(\mathbf{x}^n)]. \quad (\text{B.47})$$

The implementation forms separate normalized infinity-norm residuals for the structural increments and for the generalized aerodynamic loads. For an active degree of freedom  $j$ , the state residuals have the form

$$r_z = \max_j \frac{|\Delta z_j^{(k+1)} - \Delta z_j^{(k)}|}{\max \left( z_{\text{scale},j}, |\Delta z_j^{(k+1)}|, |\Delta z_j^{(k)}| \right)}, \quad z \in \{q, \dot{q}, \ddot{q}\}, \quad (\text{B.48})$$

and the load residual uses the analogous denominator with the configured force scale and the previous and current generalized loads. The locked flap is excluded from these active-DOF norms. The primary gate is  $r_q$ ,  $r_{\dot{q}}$ , and  $r_Q$  below the configured tolerance;  $r_{\ddot{q}}$  must also pass, except for the implemented repeated float32 round-off stagnation condition after the primary gate has passed.

The algorithm applies relaxation after the state residuals are computed and before the candidate geometry is evaluated. For a structural increment  $\Delta z$ ,

$$\Delta z \leftarrow \omega \Delta z + (1 - \omega) \Delta z_{\text{prev}}, \quad z \in \{q, \dot{q}, \ddot{q}\}, \quad (\text{B.49})$$

where  $\omega$  is fixed or computed by the Aitken update and then clamped to the configured interval. During this iteration, the accepted wake circulation is supplied to the wake-influence right-hand side; the candidate trailing-edge station is generated geometrically, but its circulation is committed only by the post-convergence call that accepts the new wake row.

### B.5 Inferred-balanced structural reconstruction

The inverse mass-property hypothesis used for the conditional  $U = 0$  check is explicitly

$$S_{\alpha,\text{base}} = 0, \quad I_{\alpha,\text{base}} = 0.014 \text{ kg m}^2. \quad (\text{B.50})$$

The 40 g point mass is located at 0.280 m from the leading edge and the elastic axis is at 0.050 m. Its contributions are therefore

$$\begin{aligned} \Delta S_{\alpha} &= 0.040(0.280 - 0.050) = 0.0092 \text{ kg m}, \\ \Delta I_{\alpha} &= 0.040(0.280 - 0.050)^2 = 0.002116 \text{ kg m}^2. \end{aligned} \quad (\text{B.51})$$

The resulting totals are  $m = 2.44 \text{ kg}$ ,  $S_{\alpha} = 0.0092 \text{ kg m}$  and  $I_{\alpha} = 0.016116 \text{ kg m}^2$ . The estimated flap properties are  $S_{\beta} = 0.0084125 \text{ kg m}$  and  $I_{\beta} = 0.000733875 \text{ kg m}^2$ . The matrices used by the independent structural check are

$$\begin{aligned} \mathbf{M}_{\text{OFF}} &= \begin{bmatrix} 2.44 & 0.0092 \\ 0.0092 & 0.016116 \end{bmatrix}, & \mathbf{K}_{\text{OFF}} &= \begin{bmatrix} 3470 & 0 \\ 0 & 13.18 \end{bmatrix}, & (\text{B.52}) \\ \mathbf{M}_{\text{ON}} &= \begin{bmatrix} 2.44 & 0.0092 & 0.0084125 \\ 0.0092 & 0.016116 & 0.0016171875 \\ 0.0084125 & 0.0016171875 & 0.000733875 \end{bmatrix}, & \mathbf{K}_{\text{ON}} &= \begin{bmatrix} 3470 & 0 & 0 \\ 0 & 13.18 & 0 \\ 0 & 0 & 1.108 \end{bmatrix}. \end{aligned} \quad (\text{B.53})$$

The damping matrix is not used to identify the undamped frequencies; its ratios remain provisional configuration values. The matrices above define the structural reconstruction used in the zero-flow comparison.

### B.6 Linear response and frequency extraction

The Fig. 7 frequency protocol uses

$$U \in \{2, 8, 12, 16\} \text{ m s}^{-1}, \quad T = 5 \text{ s}, \quad h(0) = 0.1 \text{ mm},$$

with  $\alpha(0) = 0.02^\circ$  and  $\beta(0) = 0.02^\circ$  in the ON configuration, and  $\beta(0) = 0$  in the OFF configuration. The initial transient is removed before applying a Hann window and extracting the dominant spectral peaks. The third ON peak is retained as a flap branch when its continuity and flap-coordinate participation support that interpretation. No RMS error is assigned because the article provides graphical readings rather than raw histories or tabulated frequency uncertainties; the comparisons in the main text are direct frequency differences. The ON case at  $U = 12 \text{ m s}^{-1}$  did not satisfy the FSI convergence criterion.

### B.7 Nonlinear response protocol

The experimental and numerical amplitude definitions are not identical in their available records. The article averages local-extremum amplitudes from the zero reference over 30 s after establishment of a stable LCO. The UVLM records 5 s and analyses approximately the final 2.5 s using decay or growth, envelope, energy, and convergence criteria. The numerical amplitude is the mean absolute local-extremum amplitude measured from the zero reference. All converged numerical cases remain damped, while the large-amplitude ON case at  $U = 20 \text{ m s}^{-1}$  does not converge. No numerical LCO amplitude or experimental–numerical amplitude RMS is therefore defined.

### B.8 Calibration and independence of the validation data

The Fig. 7 correction uses the generalized-load form

$$\Delta \mathbf{Q} = \mathbf{M}_c \ddot{\mathbf{q}} + U \mathbf{C}_c \dot{\mathbf{q}} + U^2 \mathbf{K}_c \mathbf{q}, \quad (\text{B.54})$$

with parameters fitted to the same Fig. 7 dataset used for comparison. The resulting agreement is therefore a calibration or sensitivity result, not an independent validation of the uncorrected UVLM. The calibrated response is not used as the final model for Figs. 5.5 and 5.6. The calibrated response is shown in Fig. B.1; it is retained here to keep its use of the validation data explicit.

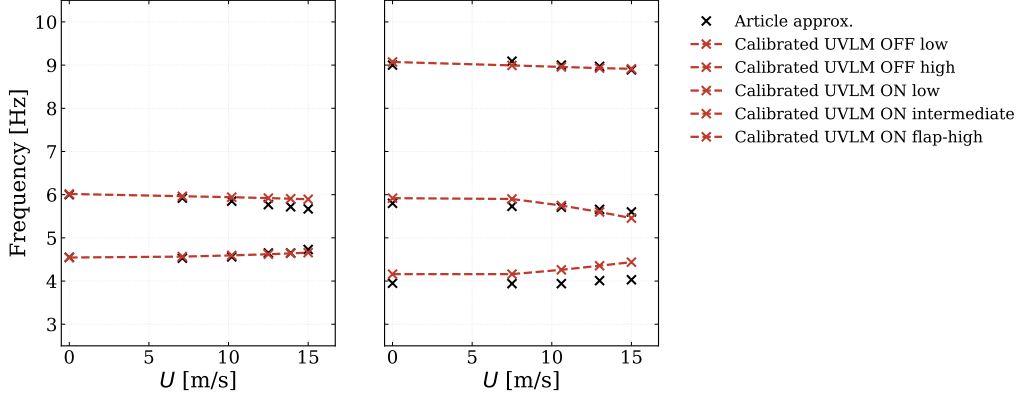


Figure B.1 Calibrated response based on Fig. 7 of García Pérez et al. obtained after fitting the generalized aerodynamic correction to the same experimental dataset. The resulting agreement is therefore a calibration result and not an independent validation of the uncorrected UVLM.

## B.9 Experimental-data and model limitations

The principal limitations of the present comparison are:

- mounted-assembly  $S_\alpha$  and  $I_\alpha$  with the flap blocked;
- complete moving-flap mass, centre of mass and inertia, including rod, shaft, encoder rotor and moving spring hardware;
- modal damping or decay measurements for the active coordinates;
- raw or tabulated Fig. 7 frequency data and Fig. 8 time histories;
- a numerical treatment of separated-flow/dynamic-stall physics if the large-amplitude Fig. 8 regime is to be compared quantitatively.